

Designing surface transport under uncertain kinetics: moment thresholds and measurement precision

7 September 2026

Abstract

Weak surface exchange can produce large variations in the spreading of tracers among nominally similar channels. We study how a fixed integral budget of tangential surface diffusivity should be distributed when the exchange pattern is uncertain, and how accurately that pattern must be measured before placement. For the dimensionless family $k_c(s) = (c + \cos s)^2$, with c uniform on $[-2, 2]$, we prove sharp small-budget asymptotics over all nonnegative integrable mobility fields. Predetermined spatial design moves the disorder-moment threshold from $q = 4/3$ for uniform mobility to $q = 8/5$, where the optimal moment is $K_{\text{crit}} M^{-2/5} [\log(1/M)]^{7/5}$. Above this threshold an attained whole-line variational problem supplies the sharp coefficient. The three optimized orders persist for smooth families with transverse folds. Exact observation changes the optimal mean from order $M^{-1/4}$ to $M^{-1/5}$. For equal observation bins of width Δ , we prove the joint law $\mathcal{Q} \asymp M^{-1/5} + M^{-1/4} \Delta^{1/4}$, an exact continuous crossover at $\Delta/M^{1/5} = O(1)$, and sharp endpoint coefficients. A bulk–surface comparison transfers these scalar optimum laws to stationary flow-induced dispersion at fixed positive bulk diffusivity. Explicit local certificates, unrestricted lower bounds, and numerical refinement distinguish asymptotic values from the performance of finite-budget trial designs.

1 Introduction

Particles transported through an adsorbing channel alternate between advection in the bulk and periods of residence at the wall. Even when the mean fraction of adsorbed particles is fixed, slow exchange at some wall locations can increase fluctuations in the time spent moving. Surface diffusion provides an escape route along the wall. The design question is therefore spatial: where should a limited amount of surface mobility be placed when the locations and strengths of slow exchange are uncertain?

Adsorption, surface motion, and their effects on dispersion are established parts of generalized Taylor dispersion theory [10, 13]. More recent formulations allow spatially varying diffusivity and wall interactions in confined microfluidic systems [2]. The present problem keeps the channel geometry, bulk flow, and equilibrium adsorption affinity fixed. The desorption rate varies along the wall, while adsorption is proportional to it with fixed proportionality. Consequently the mean speed remains fixed, and the design changes the spreading about that mean. The uncertainty is quenched: one kinetic pattern is selected for each channel and stays fixed during transport. Moments over that uncertainty describe reproducibility among channels.

Our model family on a wall of dimensionless length 2π is

$$k_c(s) = (c + \cos s)^2, \quad c \sim \text{Uniform}[-2, 2].$$

Table 1: Small-budget orders of unrooted disorder moments in the cosine ensemble. Uniform mobility has $D = M/(2\pi)$; predetermined design is optimized over every nonnegative integrable field of budget M . The threshold column separates the two open ranges in each row. All orders have sharp coefficients in the cited theorems.

Design	Threshold q_*	$q < q_*$	$q = q_*$	$q > q_*$
Uniform	4/3	$M^{-q/4}$	$M^{-1/3} \log(1/M)$	$M^{1/3-q/2}$
Predetermined	8/5	$M^{-q/4}$	$M^{-2/5} [\log(1/M)]^{7/5}$	$M^{(2-3q)/7}$

For $|c| < 1$ the rate has two quadratic zeros. They merge into a quartic zero at $c = \pm 1$ and disappear for $|c| > 1$. We refer to this merger as a *fold*. The family is deliberately simple enough to retain exact constants, while containing both ordinary kinetic defects and rare nearly merged defects. A designer chooses a nonnegative tangential diffusivity $D(s)$ with $\int_{\Gamma} D \, ds = M$. The budget is dimensionless in fixed physical units specified in [section 2](#); no powers of a dimensional budget are intended. No mobility cap or minimum fabrication length is imposed.

1.1 Results and the role of information

For a fixed pattern and mobility, the singular wall response is the variational quantity

$$J_c(D) = \sup_{v \in C^\infty(\Gamma)} \left\{ 2 \int_{\Gamma} v \, ds - \int_{\Gamma} [D|v'|^2 + k_c v^2] \, ds \right\}. \quad (1)$$

When an inverse exists this equals the integral of the solution of $-\partial_s(D\partial_s h) + k_c h = 1$. Its variational definition also allows arbitrary integrable coefficients with zero sets or fine structure. For physically admissible designs with finite response, the flow-induced dispersion equals χJ_c plus a nonnegative bulk correction, where $\chi = KV^2/(A + KP)$ is fixed by equilibrium and mean flow. We prove a logarithmic bound on that correction after adding an arbitrarily small uniform part to the mobility. This construction preserves both the budget and the information used to choose the field. It makes the leading scalar optimal values physically attainable, without taking an infinite-bulk-mixing limit.

There are two related design questions. First, a predetermined field minimizes the positive moment $\mathbb{E}[J_c(D)^q]$ before c is measured. Second, a field selected after observation minimizes the mean, with exactly budget M for every observation. The resulting moment orders are compared in [table 1](#); all their scalar coefficients are given by sharp equivalents in [theorems 3.6](#) and [4.1](#). In particular, the supercritical coefficient is an attained variational value, rather than the cost of one chosen profile.

The change of threshold has a direct interpretation. For low moments, the leading cost comes from ordinary roots moving over the wall. The formal normalized allocation is proportional to $|\sin s|^{-\alpha_q}$, where $\alpha_q = (6q - 4)/(q + 4)$. At $q = 8/5$ this shape ceases to be integrable. The actual design must then account for budget-dependent neighborhoods of the known fold sites. The proof makes this argument valid against arbitrary concentrated or oscillatory competitors. It gives the exact critical logarithm and, above the threshold, matching localization and recovery through a whole-line fold problem. The same three orders hold for a class of C^4 squared-rate families with isolated transverse folds ([theorem 5.2](#)). The generic result is an order theorem; it does not assign the cosine coefficients to other families.

Observation changes the mean more substantially than predetermined grading alone. Exact knowledge of c permits placement at the realized zeros and gives $\mathcal{O}(M) \sim K_{\text{obs}} M^{-1/5}$, whereas

$\mathcal{P}_1(M) \sim K_1 M^{-1/4}$. For equal bins of width $\Delta = 4/N$, let $\mathcal{Q}(M, \Delta)$ be the optimal mean when only the bin index is revealed. [Theorem 7.1](#) proves the uniform two-parameter law

$$\mathcal{Q}(M, \Delta) \asymp M^{-1/5} + M^{-1/4} \Delta^{1/4}, \quad M, \Delta \downarrow 0.$$

It also gives an exact crossover: $M^{1/5} \mathcal{Q}(M, \Delta) \rightarrow \mathcal{H}(\tau)$ when $\Delta/M^{1/5} \rightarrow \tau < \infty$. The continuous function $\mathcal{H}$ is an explicit integral of the attained unit-budget problem for a quadratic defect with an uncertain center. It has $\mathcal{H}(0) = K_{\text{obs}}$ and $\mathcal{H}(\tau) \sim K_{\text{coarse}} \tau^{1/4}$ at large τ ; the simultaneous coarse limit is proved separately. Thus offset resolution of order $M^{1/5}$ is the scale at which the measurement changes the leading mean performance. This is a statement about a specified noiseless quantizer, not a theory of arbitrary noisy measurements or an optimization of the sensor.

1.2 Relation to existing design and uncertainty methods

The response (1) has the variational structure of compliance. Mass-constrained reinforcement, measure relaxation, and gradient constraints are established tools in this setting [6, 8]. Deterministic coefficient design under random forcing is treated by Buttazzo and Maestre [7]; general risk criteria for mixtures of two positive conductors are studied by Alphonse et al. [3]. Those works provide close optimization precedents. The present singular limit instead concerns random vanishing reaction rates and an unrestricted nonnegative L^1 diffusivity whose total budget tends to zero. The two cited stochastic-conductivity models impose positive conductivity bounds and do not include this coefficient class. The contribution here is the sharp transport laws, including the unrestricted lower bounds and recovery constructions at the degeneracies.

Neither the relation between risk preference and resource allocation nor the importance of rare degeneracies is a new general principle. Optimized tolerance models already show that risk preference changes large-loss statistics [14]. Parameter-averaged moments can likewise select rare bifurcations because their enhanced response compensates for their small probability [5]. Here the integral mobility budget and local resolvent identify the particular thresholds, logarithm, and attained variational constants.

The information available at the design stage is also an established part of stochastic shape optimization [9]. Observation-channel continuity and quantized policies have a developed theory [15, 16]. The result proved here is a quantitative joint limit in which observation resolution and an independent, degenerating PDE budget both vary. Conditional localization must hold uniformly, and bins containing merged defects must be controlled before the local values can be averaged. These are the additional steps that produce $\mathcal{H}$ and the global resolution threshold.

1.3 Organization

[Section 2](#) derives the physical coefficient, specifies rough mobility realizations, and proves the scalar-to-bulk comparison. [Section 3](#) establishes the harmonic and fold responses, the uniform-design moments, and the limits of replacing random rates by averages or local zero statistics. [Section 4](#) proves the sharp predetermined-design theorem; [section 5](#) extends its orders to generic folds. [Section 6](#) derives the exact local placement profile and the observation-dependent mean. [Section 7](#) proves the finite-resolution theorem. [Section 8](#) supplies reproducible numerical evidence with separate discretization checks, and [section 9](#) discusses the physical scope. All arguments needed for these results are included.

2 Transport model and comparison of design problems

The wall pattern is *quenched*: it is sampled once and remains fixed while particles move. Expectations over patterns below describe variation among channels, not higher moments of the displacement of one particle. The flow and equilibrium affinity are fixed while the exchange rate and tangential surface diffusivity may vary along the wall.

2.1 Physical model and units

Let $\Omega \subset \mathbb{R}^2$ be a bounded connected cross-section with C^2 boundary Γ . Assume that Γ is connected; its arclength coordinate is periodic with period P , and $A = |\Omega|$. The bulk transverse diffusivity $D_b > 0$, affinity $K > 0$, and longitudinal velocity $u \in L^2(\Omega)$ are fixed. Adsorbed particles have zero longitudinal advective velocity. For a realization ω , the desorption rate is $k = k_\omega \geq 0$, the adsorption coefficient is Kk , and $D(s) \geq 0$ is tangential surface diffusivity. Our uniform estimates assume

$$0 \leq k \leq k_{\max}, \quad \int_{\Gamma} k \, ds \geq \kappa > 0. \quad (2)$$

When axial molecular diffusion is included, assume $0 \leq D_b^x < \infty$ and $D_s^x \in L^1(\Gamma)$ with $D_s^x \geq 0$. For smooth coefficients, bulk concentration $p(x, y, t)$ and wall concentration $\rho(x, s, t)$ satisfy

$$\begin{aligned} \partial_t p + u(y) \partial_x p &= D_b \Delta_y p + D_b^x \partial_x^2 p, \\ -D_b \partial_n p &= Kk p_{\Gamma} - k\rho, \\ \partial_t \rho &= \partial_s (D \partial_s \rho) + D_s^x(s) \partial_x^2 \rho + Kk p_{\Gamma} - k\rho. \end{aligned} \quad (3)$$

Here p_{Γ} is the bulk trace and n is the outward normal. The boundary loss in the bulk equals the wall gain. The divergence form of surface motion is part of the model; replacing it by $D \partial_s^2$ would generally change the stationary measure. The rough-coefficient interpretation is given below.

Set

$$Z = A + KP, \quad \pi(dy, \text{bulk}) = \frac{dy}{Z}, \quad \pi(ds, \text{wall}) = \frac{K \, ds}{Z}, \quad V = \frac{1}{Z} \int_{\Omega} u \, dy. \quad (4)$$

Indeed, constant bulk concentration and wall concentration K times that constant make every transverse flux vanish. The transverse dynamics are reversible with respect to π , as the symmetric form below confirms. Thus the stationary mean speed V is independent of k and D . We assume $V \neq 0$ when comparing divergent optimal values and write

$$\chi = \frac{KV^2}{Z} > 0. \quad (5)$$

Before taking a small-budget limit, all quantities must be expressed in fixed units. Dimensionally, $[k] = T^{-1}$, $[D] = L^2 T^{-1}$, $[K] = L$, and the budget $M = \int_{\Gamma} D \, ds$ has units $L^3 T^{-1}$. Choose $\ell_{\text{ref}} = P_{\text{phys}}/(2\pi)$ and a fixed reference rate $k_{\text{ref}} > 0$. Put

$$s' = s_{\text{phys}}/\ell_{\text{ref}}, \quad t' = k_{\text{ref}} t_{\text{phys}}, \quad k' = k_{\text{phys}}/k_{\text{ref}}, \quad D' = \frac{D_{\text{phys}}}{k_{\text{ref}} \ell_{\text{ref}}^2}, \quad M' = \frac{M_{\text{phys}}}{k_{\text{ref}} \ell_{\text{ref}}^3}. \quad (6)$$

Bulk lengths and K are divided by ℓ_{ref} , velocities by $k_{\text{ref}} \ell_{\text{ref}}$, and diffusivities by $k_{\text{ref}} \ell_{\text{ref}}^2$. We henceforth suppress primes and use fixed dimensionless units. We retain P in the comparison

lemmas to show their geometric dependence; the canonical ensemble has $P = 2\pi$. The surface response defined below has physical units LT and converts as

$$J_{\text{phys}} = \frac{\ell_{\text{ref}}}{k_{\text{ref}}} J', \quad D_{\text{flow,phys}} = k_{\text{ref}} \ell_{\text{ref}}^2 D'_{\text{flow}}. \quad (7)$$

In particular, a dimensionless scalar asymptotic is multiplied by $\chi_{\text{phys}} \ell_{\text{ref}} / k_{\text{ref}}$ when used for physical flow-induced diffusivity. Every subsequent power or logarithm of M refers to the dimensionless budget.

2.2 Scalar response and admissible transverse dynamics

For *every* nonnegative $D \in L^1(\Gamma)$ define the smooth-test functional

$$\mathfrak{a}_{k,D}[v] = \int_{\Gamma} D|v'|^2 ds + \int_{\Gamma} kv^2 ds, \quad v \in C^\infty(\Gamma), \quad (8)$$

$$J_k(D) = \sup_{v \in C^\infty(\Gamma)} \left\{ 2 \int_{\Gamma} v ds - \mathfrak{a}_{k,D}[v] \right\} \in (0, \infty]. \quad (9)$$

Optimization over the scalar multiple of each test gives the equivalent quotient formula

$$J_k(D) = \sup_v \frac{(\int_{\Gamma} v ds)^2}{\mathfrak{a}_{k,D}[v]}. \quad (10)$$

A zero denominator with nonzero numerator gives $+\infty$; the quotient $0/0$ is assigned zero. These definitions impose no existence of a minimizer, inverse operator, or diffusion associated with an arbitrary measurable weight.

That distinction matters: a weighted derivative preform need not be closable for every nonnegative integrable weight; the one-dimensional closability criterion is Hamza's condition [11, Theorem 3.1.6]. We call a coefficient *physically admissible* if

$$e_D[v] = \int_{\Gamma} D|v'|^2 ds, \quad v \in C^\infty(\Gamma), \quad (11)$$

is closable in $L^2(\Gamma)$. Explicitly, closability means that for every smooth sequence v_n with $v_n \rightarrow 0$ in L^2 and $e_D[v_n - v_m] \rightarrow 0$ as $n, m \rightarrow \infty$, one has $e_D[v_n] \rightarrow 0$. This condition ensures that the form-norm completion embeds unambiguously in L^2 : two energy-Cauchy approximations to the same L^2 function define the same energy and hence a unique minimal closed form. For this class we use its minimal closure, with domain $\mathcal{V}_D$, rather than choosing unspecified interface conditions at zero-mobility sets. The closure norm is $(\|v\|_2^2 + e_D[v])^{1/2}$. The closed form $\mathfrak{a}_{k,D} = e_D + \int kv^2$ has the same domain because k is bounded. No assertion of closability is needed for the larger scalar optimization class in (9).

Lemma 2.1 (Positive-background realization). *If $D \in L^1(\Gamma)$ and $D \geq m > 0$, then (11) is closable, $\mathcal{V}_D \subset H^1(\Gamma)$, and, under (2),*

$$\|v\|_2^2 \leq C \left(\|v'\|_2^2 + \int_{\Gamma} kv^2 ds \right), \quad \mathfrak{a}_{k,D}[v] \geq c \min(m, 1) \|v\|_2^2. \quad (12)$$

Consequently $h = H_{k,D}^{-1} 1 \in \mathcal{V}_D$ exists uniquely, where $H_{k,D}$ is the self-adjoint operator of the closed form, and

$$h \geq 0, \quad J_k(D) = \int_{\Gamma} h ds = \mathfrak{a}_{k,D}[h], \quad \int_{\Gamma} kh ds = P. \quad (13)$$

All constants depend only on $P, k_{\text{max}}, \kappa$.

Proof. Suppose $v_n \rightarrow 0$ in L^2 and $\sqrt{D}v'_n$ is Cauchy in L^2 . The floor implies that v'_n is Cauchy in L^2 , with limit zero by distributional differentiation. A subsequence converges pointwise almost everywhere both before and after multiplication by $\sqrt{D}$; since D is finite almost everywhere, the latter limit is also zero. This is closability, and the same argument identifies the derivative of any member of the closure with its weak H^1 derivative.

Write $\bar{v} = P^{-1} \int v$. Periodic Poincaré and the rate anchor give

$$\kappa|\bar{v}|^2 \leq 2 \int_{\Gamma} kv^2 + 2k_{\max}\|v - \bar{v}\|_2^2 \leq 2 \int_{\Gamma} kv^2 + C\|v'\|_2^2.$$

Adding the mean-zero part proves (12), first for smooth functions and then by closure. The closed-form representation theorem and this bound give the inverse and its weak equation $\mathbf{a}_{k,D}(h, v) = \int v$. The closure is stable under normal contractions: for a smooth approximation to a Lipschitz contraction T with $T(0) = 0$ and $|T'| \leq 1$, the chain rule gives $e_D[T(v)] \leq e_D[v]$ on the smooth core; approximation and weak lower semicontinuity extend the inequality to the closure. In particular $h^- \in \mathcal{V}_D$, and testing with h^- gives $-\mathbf{a}_{k,D}[h^-] = \int h^- \geq 0$, so $h^- = 0$. Testing with h and with the constant 1 proves the identities. The smooth core is dense in the form norm, so its supremum (9) equals the inverse response just obtained. $\square$

For a physically admissible D , the unnormalized transverse energy on $H^1(\Omega) \oplus \mathcal{V}_D$ is

$$\mathcal{E}_D[f, v] = D_b \int_{\Omega} |\nabla f|^2 dy + K e_D[v] + K \int_{\Gamma} k(v - f_{\Gamma})^2 ds. \quad (14)$$

The trace map $H^1(\Omega) \rightarrow H^{1/2}(\Gamma) \subset L^2(\Gamma)$ is bounded. Thus the exchange term is continuous in the product form norm; equivalently, a sequence Cauchy for $\mathcal{E}_D + \|\cdot\|_{L^2(\pi)}^2$ is Cauchy in the bulk H^1 and surface form norms and its exchange term converges. This proves closedness. The same contraction argument, including $|T(v) - T(f_{\Gamma})| \leq |v - f_{\Gamma}|$, proves the Markov property. Smooth pairs are a form core and are uniformly dense in the continuous functions on the disjoint union $\bar{\Omega} \sqcup \Gamma$. Consequently this is a regular Dirichlet form. Its normalized version $\mathcal{E}_D/Z$ defines a conservative symmetric Markov process with stationary probability π ; constants have zero energy. We use the standard regular-form/process correspondence in this step [11, Theorem 7.2.1]. For smooth coefficients, integration by parts gives bulk generator $D_b \Delta f$, wall generator $\partial_s(Dv') + k(f_{\Gamma} - v)$, and boundary condition $D_b \partial_n f = Kk(v - f_{\Gamma})$, recovering the transverse part of (3).

If $D \geq m > 0$, this process has a spectral gap and its stationary law π is ergodic. To check the needed coercivity, use the gauge $\int_{\Omega} f = 0$, obtainable by subtracting a common constant from (f, v) . Bulk Poincaré and trace estimates bound $\|f\|_{H^1}$ and $\|f_{\Gamma}\|_2$ by $C\|\nabla f\|_2$. Moreover,

$$\int_{\Gamma} kv^2 \leq 2 \int_{\Gamma} k(v - f_{\Gamma})^2 + 2k_{\max}\|f_{\Gamma}\|_2^2.$$

Together with (12), this bounds the squared product L^2 norm in this gauge by $C_m \mathcal{E}_D[f, v]$. Subtracting the π -mean can only decrease the $L^2(\pi)$ norm. Hence the energy controls the norm on the mean-zero subspace, which proves the gap and that the kernel consists exactly of constants. No such gap is asserted for all degenerate designs.

2.3 Stationary dispersion and the Schur identity

Let Y_t denote the stationary transverse process. Its axial advective velocity is $u(y)$ in the bulk and zero on the wall. The centered velocity g therefore equals $u - V$ and $-V$, respectively, and belongs

to $L^2(\pi)$ with mean zero. Let $\mathcal{A}_D$ be the nonnegative self-adjoint operator associated with $\mathcal{E}_D/Z$. For the advective displacement $X_t^{\text{adv}} = \int_0^t (g(Y_r) + V) dr$, stationarity and the semigroup identity give

$$\frac{\text{Var } X_t^{\text{adv}}}{2t} = \int_0^t (1 - r/t) \langle g, e^{-r\mathcal{A}_D} g \rangle_\pi dr. \quad (15)$$

The spectral measure ν_g of $\mathcal{A}_D$ is nonnegative, so the right-hand side increases to

$$D_{\text{flow}} = \int_{[0,\infty)} \lambda^{-1} \nu_g(d\lambda) \in [0, \infty], \quad (16)$$

where mass at zero gives $+\infty$. Indeed the scalar integrand before the limit is $\int_0^t (1 - r/t) e^{-\lambda r} dr$, increasing to $1/\lambda$ for $\lambda > 0$ and equal to $t/2$ at zero; monotone convergence applies. This proves existence of the stationary extended coefficient directly from particle displacement. It is finite under a positive mobility floor by the spectral gap. Finiteness implies $\text{Var } X_t^{\text{adv}} = 2D_{\text{flow}}t + o(t)$; no central limit theorem is needed for this assertion.

Independent axial Brownian motion with local diffusivities D_b^x and $D_s^x(s)$ adds

$$D_{\text{mol}} = \frac{AD_b^x + K \int_\Gamma D_s^x(s) ds}{Z}. \quad (17)$$

Here the axial Brownian driver is independent of the transverse process. The stated diffusivity assumptions and stationarity give expected integrated diffusivity $tD_{\text{mol}} < \infty$ on every finite interval. Thus the axial stochastic integral is square integrable, with variance $2tD_{\text{mol}}$. Moreover, conditional on the transverse path, the axial stochastic integral has mean zero, so there is no covariance with the advective displacement. If surface diffusion is isotropic, $D_s^x = D$ and its contribution is KM/Z . The optimization theorems below concern D_{flow} .

Completing squares in the spectral representation, or first applying $(\mathcal{A}_D + \epsilon)^{-1}$ and letting $\epsilon \downarrow 0$, gives

$$ZD_{\text{flow}} = \sup_{f,v} \{2\mathcal{F}(f,v) - \mathcal{E}_D[f,v]\}, \quad \mathcal{F}(f,v) = \int_\Omega (u - V)f dy - KV \int_\Gamma v ds. \quad (18)$$

The supremum is over the closed form domain, equivalently over smooth pairs. For completeness, the general identity used here is $\sup_w (2\langle g, w \rangle - \|\mathcal{A}_D^{1/2} w\|_\pi^2) = \int \lambda^{-1} d\nu_g$. If the kernel projection $g_0 = \mathbf{1}_{\{0\}}(\mathcal{A}_D)g$ is nonzero, the trials $w_n = ng_0$ have zero energy and objective $2n\|g_0\|_\pi^2 \rightarrow \infty$, agreeing with the spectral integral. Otherwise, on the positive spectral subspace the upper inequality follows from $2ab - \lambda b^2 \leq a^2/\lambda$. The form-domain trials $w_n = \mathbf{1}_{[1/n,n]}(\mathcal{A}_D)\mathcal{A}_D^{-1}g$, with the inverse interpreted only on the indicated spectral interval, give objectives $\int_{[1/n,n]} \lambda^{-1} d\nu_g$. Monotone convergence proves the reverse inequality, including a divergent positive-spectrum integral. Thus (18) is a consequence, not a definition, of the physical coefficient.

Proposition 2.2 (Surface lower bound and finite-response identity). *Every physically admissible design obeys*

$$D_{\text{flow}}(D, k) \geq \chi J_k(D), \quad (19)$$

with extended values allowed. If $D \geq m > 0$, let $h = H_{k,D}^{-1} \mathbf{1}$ and $w = kh$. Then

$$D_{\text{flow}}(D, k) = \chi J_k(D) + \mathcal{R}(D, k), \quad \mathcal{R} \geq 0, \quad (20)$$

where

$$\begin{aligned} Z\mathcal{R} &= \sup_{\int_{\Omega} f=0} \left\{ 2\mathcal{L}_D(f) - D_b \int_{\Omega} |\nabla f|^2 - K\mathcal{S}_D(f_{\Gamma}) \right\}, \\ \mathcal{L}_D(f) &= \int_{\Omega} (u - V)f - KV \int_{\Gamma} w f_{\Gamma}, \\ \mathcal{S}_D(b) &= \inf_{v \in \mathcal{V}_D} \left\{ e_D[v] + \int_{\Gamma} k(v - b)^2 \right\} \geq 0. \end{aligned} \tag{21}$$

The same identity holds for every physically admissible D for which $J_k(D) < \infty$, using the energy-space interpretation of h and w given in the proof. In that case $\mathcal{R}$ is finite, although no uniform bound is claimed without a floor.

Proof. In (18), take $f = 0$, $v = -V\phi$, and then the supremum over smooth ϕ . This proves (19) without assuming a finite inverse. For the identity, fix $f \in H^1(\Omega)$ and set $b = f_{\Gamma}$. The surface terms to be maximized are

$$K\{2\langle kb - V, v \rangle - \mathfrak{a}_{k,D}[v] - \int kb^2\}.$$

Since $kb \in L^2$ and $H_{k,D}$ is coercive, their maximum is $K\{\langle kb - V, H_{k,D}^{-1}(kb - V) \rangle - \int kb^2\}$. Expand this expression. Its constant term is $KV^2 J_k(D)$, its term linear in b is $-2KV \int wb$, and its quadratic term is $-K\mathcal{S}_D(b)$, because completing the square also yields $\mathcal{S}_D(b) = \int kb^2 - \langle kb, H_{k,D}^{-1}kb \rangle$. All quantities are finite. Finally $\int_{\Omega} (u - V) = KPV$ and $\int w = P$, so $\mathcal{L}_D$ annihilates constants, and $\mathcal{S}_D(b + a) = \mathcal{S}_D(b)$ by shifting its trial v by a . The bulk mean-zero gauge is therefore legitimate. Choosing $f = 0$ proves $\mathcal{R} \geq 0$.

For clarity, the extension to finite response without a floor does not presume an L^2 corrector. Complete smooth functions, modulo zero-energy functions, in the norm $\mathfrak{a}_{k,D}^{1/2}$ and call the resulting Hilbert space $\mathcal{W}$. By (10), the source $\ell(v) = \int v$ extends continuously to $\mathcal{W}$ exactly when $J_k(D) < \infty$. Its Riesz representative $h \in \mathcal{W}$ satisfies $\|h\|_{\mathfrak{a}}^2 = J_k(D)$. The map $v \mapsto \sqrt{k}v$ is continuous from the smooth energy space to $L^2(\Gamma)$ and thus extends to $\mathcal{W}$; write $w = \sqrt{k}(\sqrt{k}h) \in L^2$. Here $\sqrt{k}h$ denotes that extended image, not an assumed pointwise L^2 function h . Testing against the constant gives $\int w = P$. For $b \in L^2(\Gamma)$, the functional $\ell_b(v) = \int kbv$ is also continuous in this energy norm, since $|\ell_b(v)| \leq \|\sqrt{k}b\|_2 \mathfrak{a}_{k,D}[v]^{1/2}$. Let z_b be its Riesz representative. The surface supremum is now $K\{\|z_b - Vh\|_{\mathfrak{a}}^2 - \int kb^2\}$, with cross term $-2KV \int wb$. Moreover $\mathcal{S}_D(b) = \int kb^2 - \|z_b\|_{\mathfrak{a}}^2 \geq 0$; the infimum over $\mathcal{V}_D$ equals the one over smooth tests and then over $\mathcal{W}$, by their defining density. This proves the same completion without subtracting infinite energies. Finally $w \in L^2$, bulk Poincaré, and the trace estimate bound $|\mathcal{L}_D(f)| \leq C_{D,k} \|\nabla f\|_2$ in the mean-zero gauge. The supremum in (21) is therefore finite. $\square$

2.4 A logarithmic bound uniform in the mobility field

Theorem 2.3 (Finite-bulk remainder). *Under (2), if $D \in L^1(\Gamma)$ and $D \geq m > 0$, then*

$$0 \leq \mathcal{R}(D, k) \leq C\{1 + \log_+(1/m)\}. \tag{22}$$

The constant depends only on the fixed geometry, $D_b > 0$, K , $\|u\|_{L^2(\Omega)}$, $k_{\max}$, and κ . In particular it is independent of derivatives and upper bounds of D . Here $\log_+ x = \max(0, \log x)$ in the fixed units of (6).

Proof. Replace m by $\min(m, 1)$, so it suffices to take $0 < m \leq 1$. By (12) and Cauchy–Schwarz, $J = \langle 1, H_{k,D}^{-1} 1 \rangle \leq C/m$. The energy identity gives $\|h'\|_2^2 \leq J/m \leq C/m^2$. Since $h \geq 0$, its mean is J/P , and the fundamental theorem of calculus for H^1 functions on the circle gives

$$\|h\|_\infty \leq J/P + \sqrt{P}\|h'\|_2 \leq C/m, \quad 0 \leq w = kh \leq C/m, \quad \int_\Gamma w = P. \quad (23)$$

We spell out the estimate that prevents this crude supremum bound from producing a power-divergent bulk error. Normalize Fourier coefficients by $\widehat{w}_n = P^{-1} \int_0^P w(s) e^{-2\pi i n s/P} ds$ and use

$$\|z\|_{H^{-1/2}(\Gamma)}^2 = P \sum_{n \in \mathbb{Z}} (1 + (2\pi n/P)^2)^{-1/2} |\widehat{z}_n|^2.$$

Positivity and the mass identity imply $\widehat{w}_0 = 1$, $|\widehat{w}_n| \leq 1$, and Parseval gives $\sum_n |\widehat{w}_n|^2 = P^{-1} \|w\|_2^2 \leq \|w\|_\infty$. For an integer $N \geq 2$, separate the nonzero modes at N :

$$\begin{aligned} \|w - 1\|_{H^{-1/2}}^2 &\leq C \sum_{n=1}^N \frac{1}{n} + \frac{C}{N} \sum_{|n| > N} |\widehat{w}_n|^2 \\ &\leq C \left(1 + \log N + \frac{\|w\|_\infty}{N} \right). \end{aligned} \quad (24)$$

Taking $N = \lceil 2 + \|w\|_\infty \rceil$ and using (23) yields

$$\|w - 1\|_{H^{-1/2}}^2 \leq C \{1 + \log(1/m)\}. \quad (25)$$

In the gauge $\int_\Omega f = 0$, bulk Poincaré, the trace theorem, and $H^{-1/2}$ – $H^{1/2}$ duality imply

$$|\mathcal{L}_D(f)| \leq C \{1 + \|w - 1\|_{H^{-1/2}}\} \|\nabla f\|_2.$$

Indeed separate $w = 1 + (w - 1)$ in (21); both the bulk load and the integral of the trace are bounded by the bulk gradient, and the remaining pairing is bounded by the two trace-dual norms. Dropping $K\mathcal{S}_D \geq 0$ and maximizing $2ax - D_b x^2$ over $x \geq 0$ gives $Z\mathcal{R} \leq a^2/D_b$. Equation (25) proves the claim. $\square$

The connected one-dimensional wall is used both in the anchored Poincaré estimate and in the logarithmic Fourier sum. A disconnected wall would require additional componentwise anchors. This theorem is not uniform as D_b tends to zero, and it does not bound the remainder of arbitrary designs without a positive floor.

2.5 Information policies and transfer of optimal moments

Let (ω, Y) have a specified joint probability law on standard Borel spaces. The observation Y is the information available before the mobility is chosen. The map $\omega \mapsto k_\omega$ is Borel as a map into $L^1(\Gamma)$ and satisfies (2) almost surely. A scalar policy is a Borel map $y \mapsto D_y \in L^1(\Gamma)$ with $D_y \geq 0$ and $\int D_y = M$ for every y . A physical policy additionally uses physically admissible coefficients. We impose no minimizer existence and no interchange of expectation and pointwise infimum. Define

$$F_q(M; Y) = \inf_{\text{scalar policies}} \mathbb{E}[J_{k_\omega}(D_Y)^q], \quad (26)$$

$$G_q(M; Y) = \inf_{\text{physical policies}} \mathbb{E}[D_{\text{flow}}(D_Y, k_\omega)^q], \quad q > 0. \quad (27)$$

Every budget is per observation, rather than an average over observations. The joint observation law may depend on M .

These objectives are measurable. To see this without a measurable selection theorem, choose a countable family of real trigonometric polynomials with rational coefficients, dense in $C^1(\Gamma)$. For each smooth v the scalar expression in (9) is continuous in (k, D) in the $L^1 \times L^1$ topology; the countable family has the same supremum because all its integrands converge under C^1 approximation. Thus J is a jointly Borel extended function. For the physical response, choose similarly a countable C^1 -dense family in the smooth bulk-wall core. Each expression in (18) is continuous in (k, D) ; density of the core in every admissible form domain gives the same countable supremum. This defines a Borel function on the entire scalar coefficient space which agrees with the physical response whenever the form is closable. In particular, no assertion that the set of closable weights is itself Borel is required. The uniform policy is physical, and its positive-floor coercivity bound, uniform over k , shows that both infima are finite.

Theorem 2.4 (Same-budget, same-information comparison). *Let $q > 0$, $0 < M < 1/2$, and $0 < \theta < 1$. With $L_\theta(M) = 1 + \log_+(P/(\theta M))$,*

$$\chi^q F_q(M; Y) \leq G_q(M; Y). \quad (28)$$

There is a constant independent of the information law such that

$$G_q(M; Y)^{1/q} \leq \frac{\chi}{1-\theta} F_q(M; Y)^{1/q} + CL_\theta(M), \quad q \geq 1, \quad (29)$$

$$G_q(M; Y) \leq \left(\frac{\chi}{1-\theta} \right)^q F_q(M; Y) + C^q L_\theta(M)^q, \quad 0 < q < 1. \quad (30)$$

Consequently, along any sequence of budgets and observation laws for which

$$\frac{F_q(M; Y)}{[1 + \log(1/M)]^q} \rightarrow \infty, \quad (31)$$

one has $G_q(M; Y) \sim \chi^q F_q(M; Y)$ as $M \downarrow 0$.

Proof. The physical policy class is contained in the scalar class, so (19) proves (28). For any scalar policy, even one using nonclosable weights, form the same-information mixture

$$D_y^\theta = (1 - \theta)D_y + \theta M/P. \quad (32)$$

This is Borel, has exactly budget M for every y , and is physical by lemma 2.1. On smooth tests,

$$\mathbf{a}_{k, D_y^\theta} = (1 - \theta)\mathbf{a}_{k, D_y} + \theta\mathbf{a}_{k, M/P} \geq (1 - \theta)\mathbf{a}_{k, D_y}.$$

The quotient formula implies $J_k(D_y^\theta) \leq J_k(D_y)/(1 - \theta)$, including infinite values. By theorem 2.3, realization by realization,

$$D_{\text{flow}}(D_y^\theta, k) \leq \frac{\chi}{1-\theta} J_k(D_y) + CL_\theta(M). \quad (33)$$

For $q \geq 1$, apply Minkowski's inequality to this bound. For $0 < q < 1$, use $(a + b)^q \leq a^q + b^q$ for nonnegative a, b . Take a sequence of scalar policies whose expected responses approach F_q and let its approximation error tend to zero. This proves (29)–(30) without an optimal policy or any expectation–infimum interchange. Finally take $\theta = M$. Then $L_\theta(M) = O(1 + \log(1/M))$ and $(1 - \theta)^{-1} \rightarrow 1$; divide the inequalities by the scalar optimum and use (31). $\square$

Corollary 2.5 (Uniformity over observations in the cosine ensemble). *Let $\Gamma = \mathbb{R}/(2\pi\mathbb{Z})$, $k_c(s) = (c + \cos s)^2$, and c be uniform on $[-2, 2]$. For every observation law with this marginal, every fixed $q > 0$, and all sufficiently small M ,*

$$F_q(M; Y) \geq c_q M^{-q/5}, \quad (34)$$

and, uniformly over those observation laws,

$$1 \leq \frac{G_q(M; Y)}{\chi^q F_q(M; Y)} \leq \begin{cases} 1 + C_q M^{q/5} [\log(1/M)]^q, & 0 < q < 1, \\ 1 + C_q M^{1/5} \log(1/M), & q \geq 1. \end{cases} \quad (35)$$

Proof. For $|c| \leq 1/2$, select either root r of $c + \cos r = 0$ and fix a nonnegative, nonzero smooth bump ψ supported in $(-1, 1)$. For $\ell = M^{1/5}$ small, the periodic test $v(s) = \psi((s - r)/\ell)$ has $\int v = a_\psi \ell > 0$. Since cosine is Lipschitz with constant one, $k_c(s) \leq |s - r|^2$ on its support. Every budget- M field therefore satisfies

$$\int D|v'|^2 \leq \|\psi'\|_\infty^2 M/\ell^2, \quad \int k_c v^2 \leq C_\psi \ell^3.$$

The quotient (10) gives $J_c(D) \geq c\ell^2/(M/\ell^2 + \ell^3) = cM^{-1/5}$. This bound is pointwise in both c and D ; it applies to any policy, regardless of what it observes. The event $|c| \leq 1/2$ has probability $1/4$, proving (34). In this ensemble $k_c \leq 9$ and $\int k_c = 2\pi(c^2 + 1/2) \geq \pi$, so all comparison constants are uniform in c . Substitute $\theta = M$ and (34) into (29)–(30). For $q \geq 1$, raise the resulting ratio of q th roots to the fixed power q ; its excess is $O(M^{1/5} \log(1/M))$. For $q < 1$ the excess is $O(M^{q/5} \log(1/M)^q)$. The mixture error $O(M)$ is smaller in both cases. $\square$

The comparison concerns optimal *values* at identical budgets and information. It does not assert that an unregularized scalar optimizer exists or has a small bulk remainder. Its upper bound instead constructs physical policies with a vanishing fraction of the budget in a uniform background. The same conclusions hold after adding any uniformly bounded nonnegative molecular contribution, including (17) with fixed bulk molecular diffusion and isotropic wall diffusion. Additional fabrication constraints that exclude (32), a vanishing bulk diffusivity, a zero mean velocity, or a nonuniformly anchored rate family require separate analysis.

3 Local kinetic defects and the uniform-mobility baseline

Throughout this section $D(s) = \epsilon > 0$ and $J_\epsilon(c) = J_{k_c}(\epsilon)$, where

$$\Gamma = \mathbb{R}/(2\pi\mathbb{Z}), \quad k_c(s) = (c + \cos s)^2, \quad c \sim \text{Uniform}[-2, 2]. \quad (36)$$

The budget of this uniform design is $M = 2\pi\epsilon$. A simple zero of $c + \cos s$ gives a quadratic zero of the rate. At $c = \pm 1$ two such zeros merge, giving a quartic zero. We first justify the local source problems and the uniform estimates needed to average through these mergers. These are estimates of the exact scalar response; no bounded scalar error is inferred from a leading equivalent.

3.1 Whole-line source problems and interval localization

For a nonnegative confining potential W , write

$$\mathcal{J}(W) = \sup_{v \in C_c^\infty(\mathbb{R})} \left\{ 2 \int_{\mathbb{R}} v - \int_{\mathbb{R}} (|v'|^2 + Wv^2) \right\}. \quad (37)$$

The notation $\langle 1, (-\partial_x^2 + W)^{-1} 1 \rangle$ below always means this energy-dual quantity, not an $L^2(\mathbb{R})$ inverse applied to an L^2 constant. On an interval I , let $\mathcal{J}_I^D(W)$ and $\mathcal{J}_I^N(W)$ denote the same supremum over $H_0^1(I)$ and $H^1(I)$, respectively. The superscripts specify the variational spaces; the Neumann condition is natural in the weak equation.

Lemma 3.1 (Expanding intervals, including free endpoints). *For $W(x) = x^2 + z$, with z in a compact subset of $[0, \infty)$, or $W(x) = (x^2 - \mu)^2$, with μ in a compact subset of $\mathbb{R}$, the whole-line response is finite and positive. Both interval responses on $(-R, R)$ converge to it as $R \rightarrow \infty$, uniformly in the stated compact parameter sets. Their source solutions and the whole-line source solution satisfy, with $p = 2$ or 4 respectively,*

$$\int (|v'|^2 + Wv^2) \geq c \int (1 + |x|^p)v^2, \quad (38)$$

$$\int_{|x|>L} |v| \leq CL^{(1-p)/2} \left[\int (|v'|^2 + Wv^2) \right]^{1/2}, \quad L \geq L_0. \quad (39)$$

The integrals on a finite interval are restricted to that interval; (38) holds for every H^1 function there once $R \geq R_0$. The constants are uniform over each compact parameter set.

The convergence remains true along varying intervals $R_n \rightarrow \infty$ and parameters converging in those sets if the derivative, potential, and source integrals acquire positive weights uniformly bounded above and below, with the weights converging locally uniformly to one.

Proof. Choose L_0 beyond all potential zeros and large enough that $W(x) \geq c|x|^p$ for $|x| \geq L_0$, uniformly in the parameters. On the fixed core $[-L_0 - 1, L_0 + 1]$, the fundamental theorem of calculus, followed by averaging an anchor point in $[L_0, L_0 + 1]$, gives

$$\int_{-L_0}^{L_0} v^2 \leq C \left(\int_{-L_0-1}^{L_0+1} |v'|^2 + \int_{L_0}^{L_0+1} v^2 \right).$$

The anchor interval has a uniformly positive potential. The potential controls both the exterior L^2 mass and its $|x|^p$ weight. This proves (38), without endpoint conditions. Weighted Cauchy–Schwarz proves (39), since $\int_{|x|>L} |x|^{-p} dx = 2L^{1-p}/(p-1)$. It also proves $|\int v| \leq C\mathfrak{e}[v]^{1/2}$ for the indicated energy $\mathfrak{e}$. Thus the source is continuous on its Hilbert energy completion, and Riesz representation gives a unique maximizer h with $\mathfrak{e}[h] = \int h = \mathcal{J}$. It is nonnegative by replacing a trial with its absolute value. The same argument on finite intervals gives uniformly bounded responses and energies of their maximizers. A nonzero nonnegative compact test proves positivity.

Compact smooth functions are dense in the whole-line energy space: cut off first, using the integrability of v' , v , and $|x|^{p/2}v$, and then mollify on the compact support. Every such test is available on every sufficiently large Dirichlet interval. This proves the lower limit for both interval problems. For the Neumann upper limit, take a sequence of maximizers realizing the upper limit. The energy bound and (38) give a subsequence weakly convergent in H^1 and strongly convergent in L^2 on every compact interval. The limit h has no greater energy, by lower semicontinuity and then expansion of the compact interval. Equation (39) makes the source tails uniformly small, so the source integrals converge. Consequently

$$\limsup \mathcal{J}_{(-R,R)}^N(W) \leq 2 \int h - \mathfrak{e}[h] \leq \mathcal{J}(W).$$

The same proof allows converging parameters and locally converging weights: uniform comparability preserves the anchor and tail bounds, and local convergence identifies the limiting energy and source.

It proves continuity of the whole-line response by applying the same argument directly on $\mathbb{R}$. Finally, failure of uniform convergence on a compact parameter set would give a sequence with a convergent parameter subsequence contradicting this argument. $\square$

A partition of a circle or interval gives the useful bracketing rule

$$\sum_j \mathcal{J}_{I_j}^D \leq J \leq \sum_j \mathcal{J}_{I_j}^N. \quad (40)$$

For the lower bound one may retain only selected intervals and extend their zero-trace trials by zero. For the upper bound independent endpoint traces enlarge the trial space. Energies and sources are additive, so there are no interface terms. On a complementary interval where the potential is positive, discarding the derivative energy bounds its response by $\int W^{-1}$. The same enlargement bounds the original lowest Rayleigh quotient below by the minimum of the Neumann quotients of the pieces.

Proposition 3.2 (Harmonic coefficient and isolated zeros). *For $z \geq 0$,*

$$\mathcal{J}(x^2 + z) = \frac{\pi \Gamma((z+1)/4)}{2 \Gamma((z+3)/4)}, \quad C_0 := \mathcal{J}(x^2) = \frac{\pi \Gamma(1/4)}{2 \Gamma(3/4)}. \quad (41)$$

In particular $C_0 = 4.6474760094\dots$. If a fixed nonnegative C^2 rate k_0 on a compact circle has finitely many zeros s_j , at least one, all with $k_0(s_j + x) = a_j x^2 + o(x^2)$ and $a_j > 0$, then

$$J_{k_0}(\epsilon) \sim C_0 \epsilon^{-1/4} \sum_j a_j^{-3/4}. \quad (42)$$

If there are no zeros, $J_{k_0}(\epsilon) \rightarrow \int k_0^{-1}$ instead.

Proof. For completeness the oscillator kernel is

$$K_t(x, y) = \frac{1}{\sqrt{2\pi \sinh(2t)}} \exp \left[-\frac{(x^2 + y^2) \cosh(2t) - 2xy}{2 \sinh(2t)} \right].$$

Differentiating verifies $\partial_t K_t = (\partial_x^2 - x^2) K_t$; the small- t Gaussian limit is the identity kernel. Thus it gives the oscillator heat semigroup. Two Gaussian integrations yield $\int_{\mathbb{R}^2} K_t = \sqrt{2\pi / \sinh(2t)}$. First apply the L^2 resolvent identity to $\mathbf{1}_{[-L, L]}$. These sources converge to the constant in the dual energy norm by (39), and kernel positivity permits monotone convergence. Hence

$$\mathcal{J}(x^2 + z) = \int_0^\infty e^{-zt} \sqrt{\frac{2\pi}{\sinh(2t)}} dt = \frac{\sqrt{\pi}}{2} B\left(\frac{z+1}{4}, \frac{1}{2}\right),$$

where the last substitution is $r = e^{-4t}$. This proves (41) without treating 1 as an L^2 vector.

Choose disjoint fixed intervals around the zeros with $(a_j - \eta)x^2 \leq k_0(s_j + x) \leq (a_j + \eta)x^2$. Their complement has a positive lower bound for k_0 and contributes $O_\eta(1)$ to the upper bracket. Scaling $x = (\epsilon/a)^{1/4}y$ shows that either local quadratic interval response equals $\epsilon^{-1/4}a^{-3/4} \mathcal{J}_{(-R, R)}^B(y^2)$, where $R \rightarrow \infty$ and $B \in \{D, N\}$. Apply lemma 3.1, then let $\eta \downarrow 0$. This proves (42). With no zeros, the upper bound is $\int k_0^{-1}$; the trial $v = 1/k_0$, approximated if necessary in H^1 , gives the matching limit as its derivative energy tends to zero. $\square$

3.2 A merging pair of zeros

Definition 3.3. The quartic pair response is

$$\mathcal{C}_{\text{pair}}(\mu) = \mathcal{J}((x^2 - \mu)^2), \quad \mu \in \mathbb{R}. \quad (43)$$

It is a positive continuous function by [lemma 3.1](#).

For a local rate $(bx^2 - t)^2$ with $b > 0$ and constant mobility ϵ , the changes of variables

$$\ell = (\epsilon/b^2)^{1/6}, \quad \mu = \frac{t}{\epsilon^{1/3}b^{1/3}}, \quad x = \ell y \quad (44)$$

give the exact whole-line response $\epsilon^{-1/2}b^{-1}\mathcal{C}_{\text{pair}}(\mu)$. Indeed the energy scale of the operator is $\epsilon^{2/3}b^{2/3}$ and the integrated inverse scales by $\ell/(\epsilon^{2/3}b^{2/3}) = \epsilon^{-1/2}b^{-1}$.

Proposition 3.4 (Exact pair tails). *As $\mu \rightarrow +\infty$ and $a \rightarrow +\infty$, respectively,*

$$\mathcal{C}_{\text{pair}}(\mu) \sim \frac{C_0}{\sqrt{2}}\mu^{-3/4}, \quad \mathcal{C}_{\text{pair}}(-a) \sim \frac{\pi}{2}a^{-3/2}. \quad (45)$$

Consequently

$$\mathcal{C}_{\text{pair}}(\mu) \leq C \begin{cases} (1 + \mu)^{-3/4}, & \mu \geq 0, \\ (1 + |\mu|)^{-3/2}, & \mu < 0, \end{cases} \quad (46)$$

and $\int_{\mathbb{R}} \mathcal{C}_{\text{pair}}(\mu)^q d\mu < \infty$ exactly when $q > 4/3$.

Proof. For positive μ , put $r = \sqrt{\mu}$ and take neighborhoods $|x \mp r| < \eta r$, where $\eta = \mu^{-3/8}$. On each one, $(x^2 - \mu)^2 = 4\mu(x \mp r)^2[1 + O(\eta)]$, uniformly. Their harmonic interval radii after scaling are comparable to $\eta\mu^{3/4} \rightarrow \infty$. Bracketing and [lemma 3.1](#) and [proposition 3.2](#) therefore give the combined contribution $2C_0(4\mu)^{-3/4}[1 + o(1)]$. For the upper bracket the omitted pieces contribute at most

$$\int_{|x-r|, |x+r| \geq \eta r} \frac{dx}{(x^2 - \mu)^2} \leq C\mu^{-3/2}\eta^{-1} = o(\mu^{-3/4}).$$

To check this bound, set $x = \sqrt{\mu}z$; the reciprocal has integrable z^{-4} tails and only inverse-square singularities at ± 1 , cut off at distance η . Thus the positive tail is exact. For $\mu = -a$, set $x = \sqrt{a}z$. The response becomes $a^{-3/2}$ times that for $-a^{-3}\partial_z^2 + (1+z^2)^2$. Dropping the derivative term bounds the latter by $\int (1+z^2)^{-2} dz = \pi/2$. Conversely, the trial $v = (1+z^2)^{-2}$ has finite derivative energy and gives $\pi/2 - a^{-3} \int |v'|^2$; it is admissible by energy cutoff. This proves the second tail. Continuity and positivity on compact sets now prove the envelope and the integrability assertion. $\square$

The compact-parameter interval limit and these whole-line tails are distinct statements. In particular, a rootless interval with $\mu \rightarrow -\infty$ need not have the whole-line coefficient $\pi/2$ if its radius divided by $\sqrt{|\mu|}$ stays bounded. No such joint interval equivalent is used here.

3.3 Uniform estimates for the compact cosine family

Let $t = 1 - |c|$, $d = \epsilon^{1/3}$, and let $\lambda_\epsilon(c)$ be the lowest eigenvalue of $-\epsilon\partial_s^2 + k_c$ on the circle.

Lemma 3.5 (Uniform cosine localization). *There are constants independent of $|c| \leq 2$ and $0 < \epsilon \leq \epsilon_0$ such that*

$$\begin{aligned} t \geq 0: \quad & J_\epsilon(c) \leq C\epsilon^{-1/4}(t+d)^{-3/4}, \quad \lambda_\epsilon(c) \geq c_*\epsilon^{1/2}(t+d)^{1/2}, \\ t < 0: \quad & J_\epsilon(c) \leq C(|t|+d)^{-3/2}, \quad \lambda_\epsilon(c) \geq c_*(|t|+d)^2. \end{aligned} \quad (47)$$

With $b = 1/2$, uniformly for $\mu = t/(\epsilon/2)^{1/3}$ in every fixed compact set,

$$J_\epsilon(c) = 2\epsilon^{-1/2}\{\mathcal{C}_{\text{pair}}(\mu) + o(1)\}. \quad (48)$$

On the zero-containing side there is also the following separate uniform matching statement: along every sequence $|c| < 1$ for which $\epsilon/(1 - |c|)^3 \rightarrow 0$,

$$J_\epsilon(c) = 2C_0\epsilon^{-1/4}(1 - c^2)^{-3/4}[1 + o(1)]. \quad (49)$$

For fixed $|c| > 1$ the response converges to $\int k_c^{-1}$.

Proof. It suffices to treat the fold $c = 1$; translation by π treats $c = -1$. Center $x = s - \pi$ and choose a fixed small interval $|x| < x_0$. The coordinate $y = 2\sin(x/2)$ gives the exact identity

$$c + \cos s = y^2/2 - t, \quad ds = b_0(y) dy, \quad b_0(y) = (1 - y^2/4)^{-1/2}.$$

Here b_0 and b_0^{-1} are bounded and positive. The derivative energy has weight b_0^{-1} , and the source and reaction have weight b_0 . Scale $y = (4\epsilon)^{1/6}z$. For μ in a compact set the interval expands, its weights converge locally uniformly to one, and [lemma 3.1](#) applies. The complementary fixed wall interval has rate bounded away from zero and contributes $O(1)$. This proves (48), the response bound $C\epsilon^{-1/2}$, and, by the anchored inequality and Neumann bracketing, the gap bound $c_*\epsilon^{2/3}$ when $|t| \leq C_1d$ for any fixed C_1 .

For $t \geq d$ near the fold, the roots in the x coordinate are $\pm r$, where $r \asymp \sqrt{t}$. On neighborhoods $|x \mp r| < \eta_0 r$ with a sufficiently small fixed $\eta_0 > 0$, $k_c \asymp t(x \mp r)^2$; the two comparison constants are uniform. The harmonic scale is $\ell \asymp (\epsilon/t)^{1/4}$ and $r/\ell \asymp (t^3/\epsilon)^{1/4}$. When this ratio exceeds a fixed large constant, [lemma 3.1](#) bounds each Neumann response by $C\epsilon^{-1/4}t^{-3/4}$ and its Rayleigh quotient below by $c_*\sqrt{\epsilon t}$. On the remaining pieces, $k_c \geq c_*t^2$ and $\int k_c^{-1} \leq Ct^{-3/2}$; the latter is at most $C\epsilon^{-1/4}t^{-3/4}$. Also $t^2 \geq \sqrt{\epsilon t}$, as $t^3 \geq \epsilon$. The bounded intermediate ratios are already covered by the compact fold argument. This proves the inside bounds near the fold. Away from the fold the roots have uniformly positive slopes and uniformly separated neighborhoods, so the identical harmonic brackets with fixed neighborhoods prove the inside bounds throughout $0 \leq t \leq 1$.

For $t = -\nu < 0$, translation gives $k_c = (\nu + 1 - \cos s)^2 \geq \nu^2 + (1 - \cos s)^2$. The quartic gap at $\nu = 0$ thus gives $\lambda_\epsilon(c) \geq \nu^2 + c_*\epsilon^{2/3}$. When $\nu \geq d$, the direct reciprocal-rate upper bound is $\int k_c^{-1} \leq C\nu^{-3/2}$, by comparison of $1 - \cos s$ with squared distance to its minimum. For $\nu \leq d$ use the compact fold bound. This proves all of (47).

For sharp separated-root matching near a fold put $\rho = \epsilon/t^3 \rightarrow 0$, and now choose shrinking relative neighborhoods of radius ηr , with $\eta = \rho^{1/8}$. Taylor's formula at either root gives $k_c = (1 - c^2)(x \mp r)^2[1 + O(\eta)]$ uniformly there. Their scaled harmonic radii are comparable to $\eta\rho^{-1/4} = \rho^{-1/8} \rightarrow \infty$. The complementary reciprocal-rate integral is at most $Ct^{-3/2}\eta^{-1}$, whose ratio to the proposed leading response is $O(\rho^{1/4}/\eta) = O(\rho^{1/8})$. Harmonic bracketing proves (49). Sequences remaining away from the folds are handled by fixed-root localization, uniformly by compactness of their root neighborhoods. These two alternatives prove the stated sequential uniformity. Finally, for fixed $|c| > 1$ the rate is positive and the reciprocal-rate trial used in [proposition 3.2](#) proves convergence. $\square$

3.4 Positive moments without design

Theorem 3.6 (Uniform-mobility disorder moments). *For every fixed $q > 0$, as $\epsilon \downarrow 0$,*

$$\mathbb{E}[J_\epsilon(c)^q] \sim \begin{cases} \frac{2^q C_0^q}{4} B\left(\frac{1}{2}, 1 - \frac{3q}{4}\right) \epsilon^{-q/4}, & 0 < q < 4/3, \\ \frac{2^{1/3} C_0^{4/3}}{6} \epsilon^{-1/3} \log(1/\epsilon), & q = 4/3, \\ 2^{q-4/3} \epsilon^{1/3-q/2} \int_{\mathbb{R}} C_{\text{pair}}(\mu)^q d\mu, & q > 4/3. \end{cases} \quad (50)$$

In particular,

$$\mathbb{E}J_\epsilon \sim \frac{C_0^2}{\sqrt{\pi}} \epsilon^{-1/4}, \quad (51)$$

$$\text{Var}(J_\epsilon) \sim 2^{2/3} \epsilon^{-2/3} \int_{\mathbb{R}} C_{\text{pair}}(\mu)^2 d\mu. \quad (52)$$

The mean coefficient is 12.18594958... The variance coefficient in (52) is defined by its convergent integral; any decimal approximation to that integral is numerical evidence. To express every formula at budget M , substitute $\epsilon = M/(2\pi)$, including in the logarithm.

Proof. For $|c| < 1$, (49) gives the pointwise limit $\epsilon^{1/4} J_\epsilon(c) \rightarrow 2C_0(1 - c^2)^{-3/4}$; for $|c| > 1$ the limit is zero. On the inside, (47) bounds the q th power by $Ct^{-3q/4}$. On the outside, writing $\nu = |t| = dz$, the same estimates give $\epsilon^{1/4} J_\epsilon(c) \leq C\nu^{-3/4}$: indeed $z^{3/4}(1+z)^{-3/2}$ is bounded. This integrable dominating function for $q < 4/3$ proves the first line of (50); the density is $1/4$ and $\int_{-1}^1 (1 - c^2)^{-3q/4} dc = B(1/2, 1 - 3q/4)$.

For $q > 4/3$, use $t = (\epsilon/2)^{1/3} \mu$ near each of the two folds. Equation (48) and the rescaled envelope (47) permit dominated convergence on the expanding μ intervals, by proposition 3.4. Contributions outside fixed fold neighborhoods are $O(\epsilon^{-q/4} + 1)$, smaller than $\epsilon^{1/3-q/2}$. The two folds, density $1/4$, amplitude $2^q \epsilon^{-q/2}$, and Jacobian $(\epsilon/2)^{1/3}$ give $2^{q-4/3}$ exactly.

At $q = 4/3$, (49) and $1 - c^2 = t(2 - t)$ give the inside separated-root contribution near either fold as

$$\frac{1}{4} (2C_0)^{4/3} \epsilon^{-1/3} [t(2 - t)]^{-1} [1 + o(1)] dt.$$

Here is a precise logarithmic cutoff argument. Fix $0 < b < 1/3$ and a small fixed $t_0 > 0$. On $\epsilon^b \leq t \leq t_0$ the separated-root error tends uniformly to zero. Its integral at both folds, divided by $\epsilon^{-1/3} \log(1/\epsilon)$, tends to $(2C_0)^{4/3} b/4$, with the factor $(2 - t)^{-1}$ contributing its limit $1/2$ to the logarithmic coefficient. The inside omitted interval $0 \leq t \leq \epsilon^b$ is bounded, using (47), by

$$C \epsilon^{-1/3} \int_0^{\epsilon^b} (t + d)^{-1} dt \leq C \epsilon^{-1/3} \{1 + (1/3 - b) \log(1/\epsilon)\}.$$

The outside-fold integral is $O(\epsilon^{-1/3})$, since $\int_0^{t_0} (\nu + d)^{-2} d\nu \leq d^{-1}$. Regions away from folds have the same lower order. First let $\epsilon \downarrow 0$ and then $b \uparrow 1/3$. The lower bracket and the upper error above coincide in the limit, giving $(2C_0)^{4/3}/12 = 2^{1/3} C_0^{4/3}/6$. The beta-gamma identities give (51). For the variance, $(\mathbb{E}J_\epsilon)^2 = O(\epsilon^{-1/2})$ is smaller than the second moment, which has order $\epsilon^{-2/3}$. $\square$

The squared coefficient of variation therefore grows as $\epsilon^{-1/6}$ times a finite positive constant. This describes variation between kinetic realizations, not non-Gaussian displacement of a tracer.

More explicitly, the almost-sure limiting random variable $W = \lim_{\epsilon \downarrow 0} \epsilon^{1/4} J_\epsilon(c)$ has an atom of mass $1/2$ at zero, and its positive values are at least $2C_0$. For $w \geq 2C_0$, direct inversion of its expression above gives

$$\mathbb{P}(W > w) = \frac{1}{2} \left[1 - \sqrt{1 - (2C_0/w)^{4/3}} \right] \sim 2^{-2/3} C_0^{4/3} w^{-4/3}. \quad (53)$$

The nonintegrability of higher moments of W does not imply infinite moments at fixed positive ϵ ; all those moments are finite.

3.5 A uniform finite-bulk correction for this family

The logarithmic remainder theorem already transfers divergent moments of the uniform trial. The cosine family in fact has a stronger realization-uniform expansion.

Proposition 3.7 (Exchange flux and regular bulk limit). *Under the model assumptions of [section 2](#), let $h = H_{k_c, \epsilon}^{-1} 1$ and $w = k_c h$. Uniformly for $|c| \leq 2$,*

$$\|w - 1\|_{L^2(\Gamma)} \leq C\epsilon^{1/12}. \quad (54)$$

Let f_0 be the mean-zero solution of

$$-D_b \Delta f_0 = u - V \quad \text{in } \Omega, \quad D_b \partial_n f_0 = -KV \quad \text{on } \Gamma, \quad (55)$$

and set $\mathcal{R}_0 = (D_b/Z) \int_\Omega |\nabla f_0|^2$. Then

$$\sup_{|c| \leq 2} |D_{\text{flow}}(\epsilon, c) - \chi J_\epsilon(c) - \mathcal{R}_0| \leq C\epsilon^{1/12}. \quad (56)$$

Consequently every line of [\(50\)](#) transfers to $\mathbb{E}[D_{\text{flow}}^q]$ after multiplication by χ^q ; the mean and variance equivalents transfer with factors χ and χ^2 .

Proof. The smooth periodic equation $-\epsilon h'' + k_c h = 1$ gives $\int w = P$, the energy identity $\epsilon \int |h'|^2 + \int k_c h^2 = J_\epsilon$, and $\|h\|_2^2 \leq J_\epsilon/\lambda_\epsilon$. Multiply the equation by $k_c h$ and integrate by parts. After using $\int w = P$, the result is

$$\|w - 1\|_2^2 + \epsilon \int k_c |h'|^2 = \frac{\epsilon}{2} \int k_c h^2. \quad (57)$$

With $g = c + \cos s$, direct differentiation gives $k_c'' = 2(1 - c^2) + 6cg - 4g^2$. Discard its negative quadratic term and use $\int |g|h^2 \leq (\int k_c h^2 \int h^2)^{1/2}$ to obtain

$$\|w - 1\|_2^2 \leq C\epsilon J_\epsilon \left\{ \frac{(1 - c^2)_+}{\lambda_\epsilon} + \lambda_\epsilon^{-1/2} \right\}.$$

For $t \geq 0$, substitution from [\(47\)](#) bounds this by $C[\epsilon^{1/4}t(t+d)^{-5/4} + \epsilon^{1/2}(t+d)^{-1}] \leq C\epsilon^{1/6}$. For $t < 0$ the first term vanishes and the bound is $C\epsilon(|t|+d)^{-5/2} \leq C\epsilon^{1/6}$. This proves [\(54\)](#).

The compatibility of [\(55\)](#) is $\int_\Omega (u - V) = KPV$. Bulk Poincaré and the trace estimate, together with [\(54\)](#), show that the load $\mathcal{L}_\epsilon$ in [\(21\)](#) differs in the energy-dual norm from $\mathcal{L}_0(f) = \int_\Omega (u - V)f - KV \int_\Gamma f_\Gamma$ by $O(\epsilon^{1/12})$, uniformly in c . Dropping the nonnegative Schur penalty gives the upper bound $\mathcal{R} \leq \mathcal{R}_0 + C\epsilon^{1/12}$: the unpenalized supremum is the squared dual norm of its load divided by Z . On the stated C^2 domain the Neumann solution f_0 is in H^2 for $u \in L^2$ and constant boundary datum, by the standard Neumann regularity theorem [\[12, Section 2.4\]](#). Its trace has a square integrable tangential derivative. In the Schur infimum choose $v = (f_0)_\Gamma$, obtaining

$\mathcal{S}_\epsilon((f_0)_\Gamma) \leq \epsilon \int_\Gamma |\partial_s(f_0)_\Gamma|^2$. Inserting f_0 in the bulk supremum gives the matching lower bound $\mathcal{R} \geq \mathcal{R}_0 - C(\epsilon^{1/12} + \epsilon)$. This proves (56).

The correction is uniformly bounded while every scalar positive moment diverges. For $0 < q \leq 1$ use $|(a+b)^q - a^q| \leq b^q$ for $a, b \geq 0$. For $q > 1$ the difference is bounded by $C_q(J_\epsilon^{q-1} + 1)$; Hölder bounds its expectation by a lower-order quantity than $\mathbb{E}J_\epsilon^q$. For variance use the second-moment equivalent and $\mathbb{E}D_{\text{flow}} = O(\epsilon^{-1/4})$. The same statements permit an additional uniformly bounded nonnegative molecular diffusivity. $\square$

3.6 Why local zero statistics and mean rates do not suffice

Two counterexamples delimit the averaging operation. First, on the same circle take

$$g(s) = \xi_1 \cos s + \xi_2 \sin s, \quad \xi_1, \xi_2 \text{ independent } N(0, 1), \quad k = g^2. \quad (58)$$

At each point (g, g') is a nondegenerate standard Gaussian vector. Writing $R = (\xi_1^2 + \xi_2^2)^{1/2}$, there are almost surely two simple zeros with slope magnitude R . The local limiting zero sum is $2R^{-3/2}$, with finite mean because R has density $re^{-r^2/2}$. Its second moment is infinite. Nevertheless, the constant trial in (10) gives, for *every* nonnegative integrable mobility field, even chosen after observing g ,

$$J_{g^2}(D) \geq \frac{P^2}{\int g^2} = \frac{2P}{R^2}, \quad \mathbb{E}J_{g^2}(D) = \infty. \quad (59)$$

Thus a finite mean samplewise small- ϵ limit does not justify interchanging that limit with expectation. The missing control here is on nearly vanishing amplitudes of the entire rate field. This family does not satisfy the uniform anchoring assumption (2).

For context, the classical first marked Kac–Rice formula weights a zero’s slope by its magnitude [4, Theorems 6.2 and 6.4]. More precisely, suppose a stationary Gaussian process has almost surely C^1 paths, locally finite simple zeros, mean m_g , variances $\sigma_0^2 > 0$ and $\sigma_1^2 > 0$ for its value and derivative, on a neighborhood of the observation interval. These hypotheses permit the weighted-root formula with the derivative as the auxiliary Gaussian field. Bounded continuous truncations followed by monotone convergence give the identity for the negative-power marks used below:

$$\mathbb{E} \sum_{g(s_j)=0} \psi(|g'(s_j)|) = Lp_g(0)\mathbb{E}[|U|\psi(|U|)], \quad U \sim N(0, \sigma_1^2), \quad (60)$$

where the sum is on an interval of length L and $p_g(0) = e^{-m_g^2/(2\sigma_0^2)}/(\sqrt{2\pi}\sigma_0)$. Stationarity makes g and g' independent at one point. For the zero sum $S_L = \sum |g'(s_j)|^{-3/2}$, on any interval with $L > 0$, this yields

$$\mathbb{E}S_L = Lp_g(0)\sigma_1^{-1/2} \frac{2^{-1/4}\Gamma(1/4)}{\sqrt{\pi}} < \infty, \quad \mathbb{E}S_L^2 \geq Lp_g(0)\mathbb{E}|U|^{-2} = \infty.$$

Equivalently, the slope magnitude at a typical zero, sampled by zero intensity, has density f_{zero} , with

$$f_{\text{zero}}(r) = r\sigma_1^{-2}e^{-r^2/(2\sigma_1^2)}, \quad \mathbb{P}_{\text{zero}}(R^{-3/2} > w) = 1 - \exp[-w^{-4/3}/(2\sigma_1^2)].$$

The inverse-3/2 mark has finite q th moment exactly for $q < 4/3$. These are applications of the classical identity, not new Gaussian-process results. They neither control global amplitude collapse nor provide independence of nearby zeros or a stable limit.

Second, in the bounded cosine ensemble, $\mathbb{E}k_c(s) = 4/3 + \cos^2 s \geq 4/3$. Replacing the realized rate by this mean gives

$$J_{\mathbb{E}k_c}(D) \leq \int_{\Gamma} \frac{ds}{\mathbb{E}k_c(s)} \leq \frac{3P}{4} \quad (61)$$

for every $D \geq 0$, by discarding derivative energy. The actual uniform-design mean instead diverges as (51). Averaging the rate before solving the source problem therefore removes the kinetic defects responsible for the singular response.

4 Predetermined mobility: the optimized moment threshold

A predetermined design is selected before the offset c is known. Its optimal unrooted moment is

$$\mathcal{P}_q(M) = \inf_{D \in L^1(\Gamma), D \geq 0, \int_{\Gamma} D = M} \frac{1}{4} \int_{-2}^2 J_c(D)^q dc, \quad q > 0. \quad (62)$$

The admissible fields may depend on M and may have arbitrarily fine structure. In particular, none of the lower bounds below assumes a smooth limiting design. The scalar convention is (9); a degenerate coefficient is not automatically assigned a diffusion process.

Theorem 4.1 (Sharp predetermined moments). *Let $\alpha_q = (6q-4)/(q+4)$ and let C_0 be the harmonic coefficient in [proposition 3.2](#). For each fixed $q > 0$, as $M \downarrow 0$,*

$$\mathcal{P}_q(M) \sim \begin{cases} K_q M^{-q/4}, & 0 < q < 8/5, \\ K_{\text{crit}} M^{-2/5} [\log(1/M)]^{7/5}, & q = 8/5, \\ 2^{(11q-12)/7} \mathcal{S}_q M^{(2-3q)/7}, & q > 8/5, \end{cases} \quad (63)$$

where

$$K_q = 2^{q-3} C_0^q \left[2B\left(\frac{1-\alpha_q}{2}, \frac{1}{2}\right) \right]^{1+q/4}, \quad (64)$$

$$K_{\text{crit}} = \frac{(2C_0)^{8/5}}{8} \left(\frac{4}{7}\right)^{7/5}. \quad (65)$$

The finite positive constant $\mathcal{S}_q$ is the exact whole-line variational value defined in (84). Its role in (63) includes unrestricted lower and exact-budget recovery bounds; it is not an inference from scaling alone. For the corresponding full-bulk optimization with no observation, each right-hand side is multiplied by χ^q under the hypotheses of [theorem 2.4](#).

Compared with [theorem 3.6](#), spatial design moves the moment threshold from 4/3 to 8/5. The conclusion concerns a fixed positive moment order. Taking the q th root divides the budget powers by q ; it does not change the minimizing fields. Orders below one still give convex design problems, although they should not be described as conventional risk aversion.

4.1 Convexity and lower certificates without regularity assumptions

Lemma 4.2 (Convexity and symmetry). *For every $q > 0$, $D \mapsto J_c(D)^q$ is convex in the extended sense. The infimum in (62) is unchanged if designs are required to be even and π -periodic.*

Proof. The quotient formula (10) gives

$$J_c(D)^q = \sup_f \frac{|\int f|^{2q}}{(\int D|f'|^2 + \int k_c f^2)^q}.$$

Every denominator is affine in D , and $x \mapsto x^{-q}$ is convex on $(0, \infty)$, with its lower semicontinuous extension at zero. The supremum is therefore convex. Reflection preserves k_c ; translation by π replaces c by $-c$, which has the same law. Averaging over these four transformations preserves the budget and cannot increase the expected moment. $\square$

A useful lower bound follows by sliding a compact test along the possible root positions. Fix a nonnegative $\psi \in C_c^\infty(-1, 1)$ with $\int \psi > 0$. Near the fold at zero write $c = -\cos u$, and retain $u \in [r, 3r/2]$. Let I_r be a fixed enlargement of this interval, chosen inside $(r/2, 2r)$, and put $m = \int_{I_r} D$. For $0 < m \leq c_1 r^7$, choose $\ell = \delta_*(m/r^3)^{1/4}$, with fixed sufficiently small δ_* . The supports of $\psi((s-u)/\ell)$ lie in I_r , and $k_c(s) \leq Cr^2(s-u)^2$ there. Thus

$$J_c(D)^q \geq \frac{c_q \ell^{2q}}{[B(u) + Cr^2 \ell^3]^q}, \quad B(u) = \ell^{-2} \int D(s) |\psi'((s-u)/\ell)|^2 ds.$$

Fubini gives $r^{-1} \int_r^{3r/2} B(u) du \leq Cm/(r\ell)$. Jensen's inequality applies to the negative power for every $q > 0$. Since $dc = \sin u du \asymp r du$, it yields

$$\mathbb{E}[J_c(D)^q; u \in [r, 3r/2]] \geq c_q r^{2-5q/4} m^{-q/4}. \quad (66)$$

If $m = 0$, applying the same argument with widths tending to zero makes the integrated lower bound infinite. This covers that case without assuming an inverse exists for the degenerate field.

One fixed regular interval proves $\mathcal{P}_q(M) \geq c_q M^{-q/4}$. At $q = 8/5$, use geometrically separated intervals with disjoint spatial enlargements, at radii between a fixed r_* and $CM^{1/7}$. There are $N \asymp \log(1/M)$ intervals; each satisfies the small-mass condition because $m_j \leq M$. Summing (66) gives

$$\mathbb{E}[J_c(D)^{8/5}] \geq c \sum_{j=1}^N m_j^{-2/5} \geq c N^{7/5} M^{-2/5}.$$

Above the threshold, a fixed bump $\psi(s/R)$ at the fold, with $R = M^{1/7}$, suffices. For $|c+1| \leq cR^2$, its source is of order R and its derivative and reaction energies are at most CM/R^2 and CR^5 . Consequently

$$J_c(D) \geq cR^{-3}, \quad \mathbb{E}[J_c(D)^q] \geq c_q R^{2-3q} = c_q M^{(2-3q)/7}. \quad (67)$$

These elementary certificates already identify the powers and the critical logarithm from below. Sharper certificates are needed for the coefficients.

4.2 Globally admissible graded designs

Lemma 4.3 (Graded response bounds). *Let $r(s)$ be distance to $\{0, \pi\}$, let $-1 < \alpha < 6$, and set*

$$D(s) = a_R(r(s) + R)^{-\alpha}, \quad a_R \asymp R^{6+\alpha}, \quad 0 < R \ll 1. \quad (68)$$

Constants in the comparison may depend on α and the two constants in $a_R \asymp R^{6+\alpha}$. Near a fold put $t = 1 - |c|$. Uniformly over $c \in [-2, 2]$,

$$J_c(D) \leq C a_R^{-1/4} r^{-3/2+\alpha/4}, \quad t > 0, \quad r = \sqrt{t} \geq CR, \quad (69)$$

$$J_c(D) \leq CR^{-3}, \quad |t| \leq CR^2, \quad (70)$$

$$J_c(D) \leq C|t|^{-3/2}, \quad t < 0, \quad |t| \geq CR^2. \quad (71)$$

On compact simple-root parameter sets the bound is $Ca_R^{-1/4}$; on compact rootless sets it is bounded. The same bounds hold with $r(s) + R$ replaced by $|\sin s| + R$, or when the actual mobility is bounded below by a fixed positive multiple of the displayed design.

Proof. For separated roots near a fold, take intervals of radius a fixed small multiple of r . On them the mobility is comparable to $a_R r^{-\alpha}$, and the potential is bounded below by a constant times r^2 times squared distance from the root. The natural-endpoint harmonic estimate in [lemma 3.1](#) and [proposition 3.2](#), after rescaling, gives the first bound for the root intervals. Their oscillator-scaled lengths are bounded below, because $a_R/r^{6+\alpha} \leq C$; increasing the fixed threshold r/R ensures this condition. The complement contributes at most Cr^{-3} by its reciprocal potential. This is no larger than the root bound, since their ratio is at most $C(R/r)^{(6+\alpha)/4}$.

In the central window use an interval of radius $C_2 R$, large enough to contain all roots for the stated offset window. Its mobility is comparable to R^6 . Rescaling $s = Rx$ gives derivative and potential energies of order R^5 and source of order R , with a uniformly anchored quartic interval problem. Its response is $O(R^{-3})$. The complement has reciprocal-potential integral $O(R^{-3})$. On the rootless side, discard the derivative term and integrate $C(|t| + s^2)^{-2}$; the answer is $C|t|^{-3/2}$. Fixed parameter sets are handled by the same harmonic localization or by a positive potential bound. Comparability of $|\sin s|$ and $r(s)$ proves the replacement statement; monotonicity in the mobility proves the final assertion. $\square$

Choose a_R to impose the exact budget in (68). For $\alpha = \alpha_q$ use

$$R = \begin{cases} M^{1/(6+\alpha_q)}, & q < 8/5, \\ [M/\log(1/M)]^{1/7}, & q = 8/5, \\ M^{1/7}, & q > 8/5. \end{cases} \quad (72)$$

Indeed, the normalization integral is comparable respectively to 1, $\log(1/R)$, and $R^{1-\alpha_q}$, so $a_R \asymp R^{6+\alpha_q}$ in all three cases. The root contribution to the moment is bounded by

$$Ca_R^{-q/4} \int_R^{r^*} r^{1-3q/2+\alpha_q q/4} dr, \quad 2 - \frac{3q}{2} + \frac{\alpha_q q}{4} = \frac{8-5q}{q+4}. \quad (73)$$

It is bounded at its lower endpoint below $8/5$, logarithmic at $8/5$, and controlled by that endpoint above $8/5$. The core contributes $O(R^{2-3q})$; integrating (71) gives a bounded term, a logarithm, or that same power, all within the claimed orders. This proves matching upper orders in (63). For $q > 8/5$, any $1 < \alpha < 6 - 8/q$ gives the supercritical order.

4.3 Sharp coefficients below and at the threshold

We next retain the constants lost in (66). Write $\beta = q/4$ and $W_q(r) = |\sin r|^{1-3q/2}$. Let E be the paired root arcs corresponding to a compact symmetric interval of offsets inside $(-1, 1)$, and define

$$Z_E = \int_E W_q^{1/(1+\beta)}, \quad d_E(r) = Z_E^{-1} W_q(r)^{1/(1+\beta)}.$$

Extend this reference formula smoothly slightly beyond E . It specifies test functions, not a constraint on the competing design. Choose a real compact smooth ψ with positive integral and put

$$I = \int \psi, \quad T = \int |\psi'|^2, \quad U = \int x^2 \psi^2, \quad Q_\psi = T + U, \quad j_\psi = I^2/Q_\psi.$$

For a root r set $a(r) = \sin^2 r$ and

$$\begin{aligned}\ell_r &= (Md_E(r)/a(r))^{1/4}, & h_r(s) &= (Md_E(r)a(r))^{-1/2}\psi((s-r)/\ell_r), \\ z_c &= M^{-1/4}d_E(r)^{-1/4}a(r)^{-3/4}.\end{aligned}$$

The paired test h_c has source $2Iz_c$. Its reaction energy is $2Uz_c + o(z_c)$ uniformly on these offsets. Take the reference energy $E_c^0 = 2Q_\psi z_c$, without requiring it to be the energy of any single global reference field. The supporting line of x^{-q} at E_c^0 gives, for every competing D ,

$$J_c(D)^q \geq (2j_\psi)^q z_c^q \left[1 + \frac{qT}{Q_\psi} - \frac{q}{2Q_\psi z_c} \int D|h'_c|^2 + o(1) \right]. \quad (74)$$

The error is independent of D ; a negative right-hand side causes no difficulty for this supporting inequality.

For clarity, the moving-root calculation, including its probability factor, is recorded explicitly. The marked-root density is $\rho(r) = |\sin r|/4$. Summing over both roots counts each offset twice, so

$$\mathbb{E}[(2j_\psi)^q z_c^q; c \text{ retained}] = K_{\psi,E} M^{-\beta}, \quad K_{\psi,E} = 2^{q-3} j_\psi^q Z_E^{1+\beta}. \quad (75)$$

The derivative penalty in (74) has density

$$\mathcal{K}_M(s) = \mathbb{E} \left[\frac{q(2j_\psi)^q}{2Q_\psi} z_c^{q-1} |h'_c(s)|^2; c \text{ retained} \right].$$

Changing from a root position r to $y = (s-r)/\ell_r$ gives

$$\|\mathcal{K}_M\|_\infty \leq \left[\frac{qT}{Q_\psi} K_{\psi,E} + o(1) \right] M^{-1-\beta}. \quad (76)$$

Here is the calculation behind this uniform bound. The squared amplitude, squared derivative scale, and root Jacobian combine to $M^{-5/4} d_E^{-5/4} a^{-3/4}$ before multiplying by z_c^{q-1} . The resulting spatial factor is

$$M^{-1-\beta} \rho d_E^{-1-\beta} a^{-3q/4} = \frac{1}{4} M^{-1-\beta} Z_E^{1+\beta}.$$

The derivative profile integrates to T . All coefficients vary by $o(1)$ across a support, uniformly on the fixed arcs. With the prefactor in $\mathcal{K}_M$, this gives precisely (76). At an arc endpoint only part of the nonnegative kernel is retained, so the same upper bound holds even there and just outside the arc. It can therefore be integrated against an arbitrary concentrated D .

Averaging (74), using $\int D = M$, cancels the two qT/Q_ψ terms. Thus

$$\liminf_{M \downarrow 0} M^{q/4} \mathcal{P}_q(M) \geq K_{\psi,E}. \quad (77)$$

The quotient supremum of the compact tests is C_0 by [proposition 3.2](#). For $q < 8/5$, expanding E to all root arcs gives $Z_E \rightarrow \int_0^{2\pi} |\sin r|^{-\alpha_q} dr$, which equals the beta integral in (64) and is finite.

For the upper coefficient, take the exact-budget fields

$$D_M(s) = M \frac{(|\sin s| + R)^{-\alpha_q}}{\int_0^{2\pi} (|\sin v| + R)^{-\alpha_q} dv}, \quad R = M^{1/(6+\alpha_q)}. \quad (78)$$

For each fixed simple-root offset, the normalized shape converges smoothly near both roots, and harmonic bracketing gives its local coefficient $2C_0$. The graded estimates furnish the uniform envelope

$$M^{q/4} J_c(D_M)^q \leq C |1 - |c||^{-\gamma_q}, \quad \gamma_q = \frac{7q}{2(q+4)} < 1 \quad (79)$$

near either fold, apart from the single fold point. On the root side this is (69). In the core, $M \asymp R^{6+\alpha_q}$ makes its left side at most $CR^{-2\gamma_q}$, which is bounded by the displayed envelope. On the rootless side outside the core, $M \leq C|t|^{(6+\alpha_q)/2}$ and (71) give the same bound. Elsewhere the normalized moment is uniformly bounded. Dominated convergence now proves the upper coefficient (64), including $4/3 \leq q < 8/5$ where the uniform-design envelope would not suffice. Explicitly, with $d_q = |\sin s|^{-\alpha_q}/Z_q$ and $Z_q = \int_0^{2\pi} |\sin s|^{-\alpha_q} ds$, the limiting integral is

$$2^{q-3}C_0^q \int_0^{2\pi} W_q(s) d_q(s)^{-q/4} ds = 2^{q-3}C_0^q Z_q^{1+q/4} = K_q.$$

At $q_* = 8/5$, let $0 < b < 1/7$, retain root distances at least M^b from the folds, and use

$$Z_{EM} = 4b \log(1/M) + O(1), \quad d_{EM}(r) = \frac{1}{Z_{EM} |\sin r|}.$$

The same certificate is uniform on these moving arcs because

$$\sup_{r \in E_M} \frac{\ell_r}{\text{dist}(r, \{0, \pi\})} \leq C \left[\frac{M}{Z_{EM} M^{7b}} \right]^{1/4} \longrightarrow 0. \quad (80)$$

The Taylor reaction error is controlled by this ratio. The logarithmic derivatives of every coefficient in the moving-root kernel are $O(1/r)$ near a fold, so its relative coefficient and Jacobian errors are controlled by the same ratio. Endpoint truncation remains nonnegative. Therefore the cancellation in (77) holds with relative errors $o(1)$ even though Z_{EM} diverges. Take the small-budget limit first, then $j_\psi \rightarrow C_0$, then $b \uparrow 1/7$. This gives (65) from below.

For the matching upper bound put

$$L = \log(1/M), \quad R = (M/L)^{1/7}, \quad Z_R = \int_0^{2\pi} \frac{ds}{|\sin s| + R}, \quad a_R = M/Z_R, \quad D_M(s) = \frac{a_R}{|\sin s| + R}.$$

There are four one-sided approaches to the two folds, hence

$$Z_R = 4 \log(1/R) + O(1) \sim \frac{4}{7}L, \quad a_R \sim \frac{7}{4}R^7. \quad (81)$$

Retain root-to-fold distances $r \geq r_0 = RL$. Around each root use intervals of half-length $\eta |\sin r|$, $\eta = L^{-1/2}$. The mobility and potential differ from their frozen quadratic values by relative $O(\eta + R/r)$, uniformly. The oscillator width divided by r is at most $C(a_R/r_0^7)^{1/4} = O(L^{-7/4})$; thus the rescaled intervals have lengths at least of order $L^{5/4}$. The complement contributes at most $C/(\eta r^3)$, whose ratio to the main contribution is $O(L^{-5/4})$. Both sides of the bracketing therefore give the uniform equivalent

$$J_c(D_M) = [1 + o(1)] 2C_0 a_R^{-1/4} (1 - c^2)^{-5/8}, \quad |c| \leq \cos r_0.$$

Its critical moment integral is

$$[1 + o(1)] \frac{(2C_0)^{8/5}}{8} a_R^{-2/5} \int_{E(r_0)} \frac{ds}{|\sin s|} \sim \frac{(2C_0)^{8/5}}{2} a_R^{-2/5} \log(1/R). \quad (82)$$

The omitted root annuli $CR < r < RL$ cost $O(a_R^{-2/5} \log L)$ by (69); the core and the rootless sides cost $O(a_R^{-2/5})$ by (70)–(71). They are negligible. Substitution of (81) into (82) proves (65).

4.4 The supercritical whole-line problem and measure limits

For a finite nonnegative measure ν on $\mathbb{R}$, define the scalar test response and its integrated moment by

$$\mathcal{J}_\mu(\nu) = \sup_{v \in C_c^\infty(\mathbb{R})} \left\{ 2 \int v \, dx - \int |v'|^2 \, d\nu - \int (x^2 - \mu)^2 v^2 \, dx \right\}, \quad (83)$$

$$\mathcal{J}_q(\nu) = \int_{\mathbb{R}} \mathcal{J}_\mu(\nu)^q \, d\mu, \quad \mathcal{S}_q = \inf_{d \geq 0, \, d \in L^1(\mathbb{R}), \, \int d = 1} \mathcal{J}_q(d \, dx). \quad (84)$$

Measures are introduced only for compactness of design sequences. Equation (83) does not specify a stochastic process with measure-valued mobility. Measure relaxation and semicontinuity of compact-test variational values are established methods in reinforcement optimization; see, for example, Buttazzo et al. [8, Proposition 4.1]. The arguments below also address the noncompact domain, singular mass, and the integrated fold response needed here.

Lemma 4.4 (Singular mass, weak limits, and scaling). *Write $\nu = d \, dx + \nu_s$ for its Lebesgue decomposition. Then*

$$\mathcal{J}_\mu(\nu) = \mathcal{J}_\mu(d \, dx), \quad \mathcal{J}_q(\nu) = \mathcal{J}_q(d \, dx). \quad (85)$$

The response is lower semicontinuous under vague convergence of finite measures, pointwise in μ ; the integrated moment is also lower semicontinuous. For $q > 8/5$, $0 < \mathcal{S}_q < \infty$ and

$$\inf_{\int d = m} \mathcal{J}_q(d \, dx) = m^{-\beta_q} \mathcal{S}_q, \quad \beta_q = \frac{3q-2}{7} > 0, \quad m > 0. \quad (86)$$

The value at $m = 0$ is infinite.

Proof. The inequality $\mathcal{J}_\mu(\nu) \leq \mathcal{J}_\mu(d \, dx)$ is immediate. For the reverse inequality fix a compact smooth v . Inside a common compact interval take compact sets K_n supporting all but $o(1)$ of the singular mass relevant to v' . Choose open neighborhoods U_n with $|U_n| \rightarrow 0$ and $\int_{U_n} d \rightarrow 0$. There are smooth $0 \leq \rho_n \leq 1$, zero near K_n and one outside U_n . Fix a smooth compact ζ of integral one and set

$$a_n = \int \rho_n v', \quad v'_n = \rho_n v' - a_n \zeta, \quad v_n(x) = \int_{-\infty}^x v'_n(y) \, dy.$$

Their supports lie in a common compact set, $a_n = O(|U_n|)$, and $v_n \rightarrow v$ uniformly. The absolutely continuous derivative energies converge to $\int d|v'|^2$, while the singular derivative energies tend to zero: the uncorrected derivative vanishes on K_n , the remaining singular mass tends to zero, and the correction has size a_n . The source and reaction integrals converge uniformly on the common compact support. This proves (85) by taking the test supremum.

For a fixed test the derivative integral is continuous under vague convergence. The supremum is therefore lower semicontinuous. The response is Borel in μ (it is a supremum of continuous functions), so Fatou proves the integrated assertion. A fixed nonnegative bump on $[-1, 1]$, optimized in amplitude, gives a positive response bound for $|\mu| \leq 1$ uniformly when $\nu(\mathbb{R}) \leq 1$. Thus $\mathcal{S}_q > 0$.

For finiteness choose

$$1 < \alpha < \min\{2, 6 - 8/q\}, \quad \rho_\alpha(x) = c_\alpha(1 + |x|)^{-\alpha}, \quad \int \rho_\alpha = 1. \quad (87)$$

The variable-mobility version of the bracketing proof in lemma 4.3 gives

$$\mathcal{J}_\mu(\rho_\alpha) \leq C \begin{cases} (1 + \mu)^{-(6-\alpha)/8}, & \mu \geq 0, \\ (1 + |\mu|)^{-3/2}, & \mu < 0. \end{cases} \quad (88)$$

For completeness, at large positive μ take intervals of radius $c\sqrt{\mu}$ around $\pm\sqrt{\mu}$. There the mobility is comparable to $\mu^{-\alpha/2}$ and the quadratic coefficient is comparable to μ , giving the displayed harmonic power. The complementary reciprocal potential is $O(\mu^{-3/2})$. For bounded μ , a fixed core has a positive mobility lower bound and an anchored energy; for negative μ use the reciprocal potential directly outside a fixed compact set of parameters. Both powers in (88) have integrable q th powers by (87) and $q > 8/5$.

Finally, if $d_m(x) = m^{6/7}d(x/m^{1/7})$, then $\int d_m = m \int d$ and a change of variables in the test supremum gives

$$\mathcal{J}_\mu(d_m dx) = m^{-3/7} \mathcal{J}_{\mu/m^{2/7}}(d dx).$$

Integration in μ proves (86) in both directions. At zero mass the response is infinite for every $\mu > 0$, by bumps at a root of $(x^2 - \mu)^2$ with widths tending to zero. $\square$

In particular, relaxation to finite measures of mass at most one does not reduce $\mathcal{S}_q$. Singular mass has no benefit for this scalar convention, and loss of mass at infinity increases the optimal value through (86). The statement also implies existence of at least one unit-mass density minimizing (84): take a minimizing sequence of measures, extract a vaguely convergent subsequence, use lower semicontinuity and (85), and then use the strict decrease of $m^{-\beta_q}$ to rule out lost or singular mass. No uniqueness, regularity, or explicit formula for that minimizer is asserted.

Lemma 4.5 (Natural endpoints with a graded positive background). *Suppose $d \in L^1(\mathbb{R})$ and $d(x) \geq a(1 + |x|)^{-\alpha}$, where $a > 0$ and $1 < \alpha < 2$. For μ in a compact set, the whole-line response is finite. On intervals $(-T_n, T_n)$ with $T_n \rightarrow \infty$, the free-endpoint response converges to $\mathcal{J}_\mu(d dx)$. The conclusion holds along converging μ_n and when the source and potential are multiplied by weights w_n that are uniformly bounded above and below and converge locally uniformly to one; the derivative energy remains $\int d|v'|^2$.*

Proof. On a fixed core the lower bound on d and a positive-potential anchor give an H^1 bound from the energy. Outside that core the potential is at least cx^4 , uniformly in the parameters. Hence $|\int w_n v| \leq C\epsilon_n[v]^{1/2}$ and

$$\int_{L < |x| < T_n} |w_n v| dx \leq CL^{-3/2}\epsilon_n[v]^{1/2}.$$

This gives uniformly bounded maximizing energies and responses. The finite-interval energy space consists of absolutely continuous functions with finite weighted derivative energy. Smooth functions are dense there: approximate the derivative in $L^2(d dx)$ by smooth functions, correct its integral with a fixed smooth bump when zero traces are required, and integrate. On compact intervals the positive lower bound on d makes derivative convergence imply uniform convergence of the anchored functions. Thus the reaction and source terms converge as well.

A sequence of maximizing functions has a locally weak H^1 limit v and locally weak weighted derivatives. Distributional derivatives identify these limits, and lower semicontinuity bounds the limiting energy. The source tails above make the source integrals converge. To use v in the whole-line smooth-test supremum, its membership in that energy completion must also be checked. For $|x|$ large, the one-dimensional Sobolev estimate on $[x - 1, x + 1]$ gives

$$|v(x)|^2 \leq C \left[|x|^{-4} + |x|^{\alpha/2-2} \right] \int_{\mathbb{R}} \{d|v'|^2 + (y^2 - \mu)^2 v^2\} dy.$$

Indeed the local L^2 norm is bounded by $C|x|^{-4}$ times the energy, and the local derivative norm by $C|x|^\alpha$ times the energy. Their product gives the second term. In particular v is bounded. Cutting off at R therefore adds derivative error at most $CR^{-2}\|v\|_\infty^2 \int_{R < |x| < 2R} d$, tending to zero; the original

energy tails also vanish. The compact derivative approximation just described then gives compact smooth convergence in energy. The limiting variational value is at most the whole-line response. Conversely, every compact smooth test is eventually an admissible interval test and its weighted integrals converge. This proves both inequalities and, by subsequences, the assertions for varying parameters and weights. $\square$

4.5 Sharp supercritical localization and exact-budget recovery

Put $b_0 = 1/2$, the coefficient in $1 - \cos s = b_0 s^2 + O(s^4)$, and first use the common scale $\ell = (M/b_0^2)^{1/7}$. Given any sequence of designs of budget M , first pass to a subsequence realizing the lower limit of its scaled objective. If that limit is infinite there is nothing to prove. Otherwise restrict this subsequence to the two cells $|s - s_j| < \pi/2$, $s_j = 0, \pi$, and push the mobility measures to $x = (s - s_j)/\ell$, dividing their masses by M . Along a further subsequence these measures converge vaguely to ν_0, ν_1 , with total masses $\theta_0 + \theta_1 \leq 1$. This compactness follows by weak compactness on each compact interval and a diagonal subsequence; total mass is bounded by one.

For offsets $c = -1 + b_0 \ell^2 \mu$ near the first fold, or $c = 1 - b_0 \ell^2 \mu$ near the second, take compact tests $h(s) = b_0^{-2} \ell^{-4} v((s - s_j)/\ell)$. Their rescaled potential tends locally uniformly to $(x^2 - \mu)^2$. The scaled source and derivative integrals therefore show

$$\liminf b_0^2 \ell^3 J_c(D_M) \geq \mathcal{J}_\mu(\nu_j). \quad (89)$$

One may first use any one compact test and then take its supremum; no exchange of an optimizer with the limit is needed. The two parameter windows are disjoint for every fixed bounded μ range once M is small. Fatou on those windows, followed by expansion of the range to $\mathbb{R}$, gives

$$\begin{aligned} \liminf M^{\beta_q} \mathbb{E}[J_c(D_M)^q] &\geq \frac{1}{4} b_0^{(3-8q)/7} [\mathcal{J}_q(\nu_0) + \mathcal{J}_q(\nu_1)] \\ &\geq \frac{1}{4} b_0^{(3-8q)/7} \mathcal{S}_q[\theta_0^{-\beta_q} + \theta_1^{-\beta_q}] \\ &\geq 2^{(11q-12)/7} \mathcal{S}_q. \end{aligned} \quad (90)$$

If a limiting measure has a singular part, its absolutely continuous mass is no greater than θ_j , so the middle inequality still follows from [lemma 4.4](#). A zero mass is interpreted as infinite cost. The final minimization uses $\theta_0 + \theta_1 \leq 1$ and the strict convexity and decrease of $\theta^{-\beta_q}$: the best allocation is $\theta_0 = \theta_1 = 1/2$. This proves the lower bound for all competitors, including sequences with concentration, oscillation, or mass escaping the rescaled fold neighborhoods.

Here is a matching recovery construction for the same unrestricted value $\mathcal{S}_q$. Take a unit-mass d with finite cost arbitrarily close to $\mathcal{S}_q$, and add $\varepsilon \rho_\alpha$ with (87). Monotonicity cannot increase the cost. Rescale spatially by the mass transformation in (86) to restore unit mass. The resulting profile, denoted $\tilde{d}$, satisfies

$$\int \tilde{d} = 1, \quad \tilde{d}(x) \geq a_\varepsilon (1 + |x|)^{-\alpha}, \quad \mathcal{J}_q(\tilde{d} dx) \leq (1 + \varepsilon)^{\beta_q} \mathcal{J}_q(d dx). \quad (91)$$

It may be averaged with its reflection, preserving these properties and decreasing the cost by convexity. No regularity or existence of a minimizing profile is needed for this construction.

Now give each fold mass $m = M/2$, put $\ell = (m/b_0^2)^{1/7}$, and on its cell use the exact coordinate

$$x = \frac{2 \sin((s - s_j)/2)}{\ell}, \quad T_\ell = \frac{\sqrt{2}}{\ell}, \quad w_\ell(x) = (1 - \ell^2 x^2/4)^{-1/2}, \quad ds = \ell w_\ell(x) dx.$$

The metric obeys $1 \leq w_\ell \leq \sqrt{2}$ on the entire cell and converges locally uniformly to one. Define the preliminary field

$$\widehat{D}_M(s) = b_0^2 \ell^6 w_\ell(x) \widetilde{d}(x) \quad \text{on each cell.} \quad (92)$$

Its mass in each cell is $m \int_{-T_\ell}^{T_\ell} w_\ell^2 \widetilde{d} dx = m[1 + o(1)]$ by dominated convergence. Multiply the whole field by $M / \int \widehat{D}_M = 1 + o(1)$ to obtain an exactly admissible D_M . This normalization changes each response by a factor between 1 and $1 + o(1)$ in whichever direction it acts: for a multiplier λ , the full quadratic energy is bounded between $\min(1, \lambda)$ and $\max(1, \lambda)$ times its old value.

Near the first fold, $c = -1 + b_0 \ell^2 \mu$ gives the exact potential $b_0^2 \ell^4 (x^2 - \mu)^2$. The scaled derivative energy from (92) is precisely $\int \widetilde{d} |v'|^2$; the source and reaction acquire the weight w_ℓ . Bracketing into the two cells and applying lemma 4.5 gives, for every fixed μ ,

$$b_0^2 \ell^3 J_{-1+b_0 \ell^2 \mu}(D_M) \longrightarrow \mathcal{J}_\mu(\widetilde{d} dx). \quad (93)$$

The other cell has a potential bounded below for these offsets and contributes $O(1)$ before rescaling. Compact supported tests give the reverse bracket. The same argument holds at the second fold.

Pointwise localization is not by itself sufficient for the moment. The lower background in (91) implies on the whole circle

$$D_M(s) \geq c_\varepsilon \ell^{6+\alpha} (r(s) + \ell)^{-\alpha}.$$

Apply lemma 4.3, using each fold for its parameter half $c \leq 0$ or $c \geq 0$. With μ the corresponding signed scaled offset, it yields the uniform envelope

$$b_0^2 \ell^3 J_c(D_M) \leq C_\varepsilon \begin{cases} (1 + \mu)^{-(6-\alpha)/8}, & \mu \geq 0, \\ (1 + |\mu|)^{-3/2}, & \mu < 0. \end{cases} \quad (94)$$

For offsets bounded away from a fold but with roots, the first bound is the regular-root bound: $\mu \asymp \ell^{-2}$, so both scales are $\ell^{(6-\alpha)/4}$. For fixed rootless offsets the second bound is the reciprocal-potential bound. Thus the envelope covers the entire half of the offset range, including its outer end; extend its left side by zero outside that range. Its q th power is integrable by (87). Dominated convergence in (93) proves

$$\lim_{M \downarrow 0} M^{\beta_q} \mathbb{E}[J_c(D_M)^q] = 2^{(11q-12)/7} \mathcal{J}_q(\widetilde{d} dx).$$

Finally choose d arbitrarily close to the infimum and then let $\varepsilon \downarrow 0$ in (91). Together with (90), this proves the supercritical equivalent in theorem 4.1.

4.6 Physical transfer and the limits of the design statements

All the moment values in (63) diverge faster than a fixed power of $\log(1/M)$. The same-budget theorem theorem 2.4 therefore transfers their exact equivalents to the physical full-bulk optimum, with coefficient χ^q . The graded trials have a positive background for each fixed M . The whole-line recovery profiles also have a positive background after transplantation to the compact wall. Alternatively, the positive-background mixture in (32) supplies admissible physical trials without changing any leading value. These statements hold at fixed positive bulk diffusivity and nonzero mean speed; no additional joint vanishing-bulk limit is asserted.

The asymptotic designs are not certified finite-budget optimizers. Nor does achieving the correct order force every design to concentrate all mobility near the folds: a successful graded trial with budget $M/2$, plus a uniform field of budget $M/2$, still achieves the supercritical order by monotonicity. The sharp supercritical lower bound instead identifies the mass allocation of asymptotically

optimal sequences at the level of the value: equality requires full mass at the two rescaled folds and equal limiting mass at them. It does not prove uniqueness or a closed form for the local minimizing density. A prescribed mobility cap, fixed minimum background, or minimum fabrication length changes the admissible small-budget problem and is outside [theorem 4.1](#).

5 Geometric robustness: smooth families with transverse folds

The optimized threshold does not require the cosine symmetry. This section proves the same three moment orders for smooth one-parameter kinetic families. The hypotheses allow unequal root slopes and additional simple roots, including roots that do not move with the parameter. The conclusion concerns orders; the sharp cosine coefficients in [theorem 4.1](#) are not assigned to other families.

Assumption 5.1 (A compact fold family). Let Γ be a connected closed wall parametrized by arclength, let I be a compact parameter interval, and let $g \in C^4(\Gamma \times I)$. The rates are $k_c(s) = g(s, c)^2$. The set

$$\{(s, c) : g(s, c) = g_s(s, c) = 0\} = \{(s_j, c_j) : 1 \leq j \leq n_f\}$$

is finite and nonempty, and every member satisfies

$$c_j \in \text{int } I, \quad g_{ss}(s_j, c_j) \neq 0, \quad g_c(s_j, c_j) \neq 0.$$

The sites s_j are pairwise distinct, as are the parameters c_j . The parameter has a probability density $\varpi \in L^\infty(I)$. In a two-sided neighborhood of each c_j , the density is bounded below almost everywhere by a positive constant.

All zeros other than the listed folds are simple in space. In particular, no realization vanishes identically. Distinct sites and parameters are explicit sufficient hypotheses: they permit disjoint spatial and parameter neighborhoods in the proof. We do not assert a simultaneous-fold extension here. A smooth density positive at every fold is an example of the sampling condition; positivity away from the folds is unnecessary.

Write $J_c^g(D) = J_{g(\cdot, c)^2}(D)$ using the smooth-test convention (9), and define the predetermined value

$$\mathcal{P}_q^g(M) = \inf_{D \in L^1(\Gamma), D \geq 0, \int_\Gamma D = M} \int_I J_c^g(D)^q \varpi(c) \, dc. \quad (95)$$

The family and $q > 0$ are fixed as the dimensionless budget tends to zero.

Theorem 5.2 (Generic optimized moment orders). *Under [assumption 5.1](#),*

$$\mathcal{P}_q^g(M) \asymp \begin{cases} M^{-q/4}, & 0 < q < 8/5, \\ M^{-2/5} [\log(1/M)]^{7/5}, & q = 8/5, \\ M^{(2-3q)/7}, & q > 8/5. \end{cases} \quad (96)$$

The constants may depend on the family, density, wall, and fixed moment order, but not on M or on a competing mobility. The lower bounds hold over all nonnegative integrable fields. Bounded fields with a strictly positive lower bound attain each upper order.

5.1 Uniform geometry and fold coordinates

We first verify the geometric facts needed to apply the local estimates uniformly. At a simple zero, choose a product neighborhood where g_s has fixed sign and is bounded away from zero. At a fold, choose one where g_{ss} has fixed sign and is bounded away from zero. A spatial slice of the first neighborhood has at most one zero, and a slice of the second has at most two, by strict monotonicity or strict convexity (concavity). A finite cover of the compact zero set therefore bounds the number of spatial zeros uniformly in c . On the closed part of this set outside open fold neighborhoods, $|g_s|$ has a positive minimum. Continuity also gives

$$\sup_{s,c} k_c(s) < \infty, \quad \inf_{c \in I} \int_{\Gamma} k_c(s) ds > 0. \quad (97)$$

Indeed, a zero minimum of the latter integral would give an identically zero realization, contradicting [assumption 5.1](#).

For clarity, a fold coordinate can be constructed directly. Near (s_j, c_j) the implicit function theorem gives a critical point $z_j(c)$ with $g_s(z_j(c), c) = 0$ and $z_j(c_j) = s_j$. Taylor's formula yields

$$g(s, c) = h_j(c) + (s - z_j(c))^2 b_j(s, c), \quad b_j(s, c) = \int_0^1 (1 - \theta) g_{ss}(z_j(c) + \theta(s - z_j(c)), c) d\theta,$$

where $h_j(c) = g(z_j(c), c)$. After shrinking the neighborhood, b_j has fixed sign σ_j and its absolute value is bounded above and below by positive constants. The C^2 coordinate $x = (s - z_j(c))\sqrt{|b_j(s, c)|}$ has a spatial Jacobian bounded above and below, and

$$g(s, c) = \sigma_j(x^2 - t), \quad t = -\sigma_j h_j(c), \quad t(c_j) = 0, \quad t'(c_j) = -\sigma_j g_c(s_j, c_j) \neq 0. \quad (98)$$

Thus $|t| \asymp |c - c_j|$, $|dc/dt| \asymp 1$, and $z_j(c) - s_j = O(|t|)$. No higher regularity of the normal form is used.

On the two-root side $t > 0$, put $r = \sqrt{t}$. Each root has distance comparable to r from the fold site, their separation is comparable to r , and their slopes have magnitude comparable to r . If u is either root, then on $|s - u| \leq \eta r$, with one fixed small $\eta > 0$,

$$k_c(s) \asymp r^2(s - u)^2. \quad (99)$$

Alternatively, $g_c \neq 0$ writes the zero curve as $c = C_j(u)$, with

$$C'_j(s_j) = 0, \quad C''_j(s_j) = -\frac{g_{ss}(s_j, c_j)}{g_c(s_j, c_j)} \neq 0.$$

Hence $|C'_j(u)| \asymp |u - s_j|$. A root-position interval of length comparable to r at distance r from the site has parameter probability comparable to r^2 .

Choose disjoint spatial neighborhoods of the fold sites and disjoint parameter neighborhoods of the fold values. They may be shrunk so that the unfolding roots remain inside the retained spatial neighborhood whenever its parameter neighborhood is active. Outside these fold products, the compact simple-zero set has uniformly nonzero slopes. It therefore admits root intervals of a fixed small radius with quadratic potential bounds and a uniform bound on their number. These intervals can be taken disjoint after shrinking: otherwise a sequence of distinct ordinary roots with separation tending to zero would limit to an excluded multiple zero. On the remaining pieces, outside the active fold neighborhood, the potential has a uniform positive lower bound. These facts provide finite partitions for Neumann bracketing. They impose no condition on g_c at ordinary roots.

5.2 Lower bounds from one sampled fold

Fix one fold and one side of its zero curve. Set $s_j = 0$ in a local arclength coordinate. For $u \in [r, 3r/2]$ let I_r be a fixed enlargement contained in $(r/2, 2r)$, and let $m = \int_{I_r} D$. The bounds on C'_j and the density give

$$\varpi(C_j(u))|C'_j(u)| \asymp r \quad \text{for almost every } u \in [r, 3r/2].$$

Take the nonnegative compact bump used in (66). If $0 < m \leq c_* r^7$, choose $\ell = \delta_*(m/r^3)^{1/4}$, with fixed sufficiently small constants c_* and δ_* . All translated supports then lie in I_r and satisfy (99). Writing

$$B(u) = \ell^{-2} \int D(s) |\psi'((s-u)/\ell)|^2 ds, \quad \frac{1}{r} \int_r^{3r/2} B(u) du \leq \frac{Cm}{r\ell},$$

the quotient formula and Jensen's inequality for the convex function $x \mapsto x^{-q}$ give

$$\begin{aligned} \int_{C_j([r, 3r/2])} J_c^g(D)^q \varpi(c) dc &\geq c_q r^2 \ell^{2q} \left[\frac{m}{r\ell} + r^2 \ell^3 \right]^{-q} \\ &\geq c_q r^{2-5q/4} m^{-q/4}. \end{aligned} \quad (100)$$

If $m = 0$, the same bumps with widths tending to zero give an infinite lower bound. This argument uses only local mass and remains valid for arbitrarily concentrated or discontinuous mobility.

For $q < 8/5$ one fixed shell and $m \leq M$ give $c_q M^{-q/4}$. At $q = 8/5$, choose a geometric sequence of shells between a fixed small radius and $CM^{1/7}$, with scale ratio large enough that both their spatial enlargements and their parameter images are disjoint. The constant C ensures $m_n \leq M \leq c_* r_n^7$ for every retained shell. There are $N \asymp \log(1/M)$ of them, and

$$\int_I J_c^g(D)^{8/5} \varpi(c) dc \geq c \sum_{n=1}^N m_n^{-2/5} \geq c N^{7/5} M^{-2/5}. \quad (101)$$

The last step uses $\sum m_n \leq M$ and convexity of the negative power.

For $q > 8/5$, put $R = M^{1/7}$ and use a bump of width R centered at s_j . For $|c - c_j| \leq cR^2$, Taylor's formula gives $k_c \leq CR^4$ on its support. The derivative energy is at most CM/R^2 , the reaction energy at most CR^5 , and the squared source is comparable to R^2 . Consequently $J_c^g(D) \geq cR^{-3}$ throughout a parameter window of probability at least cR^2 . This proves the lower order $R^{2-3q} = M^{(2-3q)/7}$. Additional roots cannot weaken any of these local lower tests.

5.3 A single design protecting all possible roots

Let $\delta_F(s)$ be arclength distance to the finite set of fold sites, and set

$$\alpha = \max \left\{ 0, \frac{6q-4}{q+4} \right\}, \quad D_M(s) = a_R (\delta_F(s) + R)^{-\alpha}, \quad a_R = \frac{M}{\int_{\Gamma} (\delta_F + R)^{-\alpha} ds}. \quad (102)$$

Choose R by (72), replacing α_q by α in its subcritical case. The normalization near the finitely many sites gives

$$a_R \asymp \begin{cases} M, & \alpha < 1, \\ M/\log(1/M), & \alpha = 1, \\ MR^{\alpha-1}, & \alpha > 1, \end{cases} \quad a_R \asymp R^{6+\alpha}. \quad (103)$$

For each $M > 0$ this field is bounded and bounded below by $ca_R > 0$. Using a nonnegative grading exponent matters: an ordinary root at one parameter value can occupy a fold site associated with another value. A negative exponent would reduce its local mobility. The choice $\alpha = 0$ for $q \leq 2/3$ protects such roots and suffices for the stated orders.

Here are the uniform response estimates for this design. In a parameter neighborhood of a fold, use the signed t in (98). There are fixed thresholds such that

$$J_c^g(D_M) \leq C[1 + a_R^{-1/4} + a_R^{-1/4} r^{-3/2+\alpha/4}], \quad t > 0, \quad r = \sqrt{t} \geq CR, \quad (104)$$

$$J_c^g(D_M) \leq C[1 + a_R^{-1/4} + R^{-3}], \quad |t| \leq CR^2, \quad (105)$$

$$J_c^g(D_M) \leq C[1 + a_R^{-1/4} + |t|^{-3/2}], \quad t < 0, \quad |t| \geq CR^2. \quad (106)$$

The constant in the core threshold can be increased to cover the transition between these regions. Away from all fold parameter neighborhoods the bound is $C(1 + a_R^{-1/4})$.

We justify these estimates, including their endpoint uniformity. On an interval centered at a root, suppose the mobility is at least d and the potential is at least $a(s - u)^2$. If its half-width is at least a fixed positive multiple of $\ell = (d/a)^{1/4}$, rescaling and Neumann bracketing give

$$J_{\text{interval}}^N \leq Cd^{-1/4} a^{-3/4}. \quad (107)$$

The rescaled Neumann harmonic response is uniformly bounded as the interval grows: retain a fixed central interval, whose derivative plus quadratic reaction energy controls constants and hence has positive coercivity, and integrate x^{-2} on the two tails. Rescaled lengths in a bounded range bounded away from zero have the same bound by coercivity and compactness. This is the order version of [lemma 3.1](#) and [proposition 3.2](#).

For a fold-side root with $r \geq CR$, take an interval of radius ηr . There $D_M \asymp a_R r^{-\alpha}$ and $k_c \asymp r^2(s - u)^2$, so its oscillator length divided by r is

$$\asymp \left(\frac{a_R}{r^{6+\alpha}} \right)^{1/4} \asymp \left(\frac{R}{r} \right)^{(6+\alpha)/4}.$$

Equation (107) bounds the two root contributions by $Ca_R^{-1/4} r^{-3/2+\alpha/4}$. Outside these root intervals but within the fold neighborhood, the normal form bounds the reciprocal-potential integral by Cr^{-3} : scale x by r and exclude fixed neighborhoods of $x/r = \pm 1$. Its ratio to the root bound is at most $C(R/r)^{(6+\alpha)/4}$, so it is absorbed.

In the central window $|t| \leq CR^2$, bracket a spatial interval $|s - s_j| \leq KR$ with fixed sufficiently large K . On this interval $D_M \asymp R^6$. With $s - s_j = Ry$ and $\tau = (c - c_j)/R^2$, the reaction divided by R^4 differs uniformly by $o(1)$ from $(\gamma_j y^2 + \beta_j \tau)^2$, where $\gamma_j = \frac{1}{2}g_{ss}(s_j, c_j)$, $\beta_j = g_c(s_j, c_j)$, and τ lies in a fixed compact interval. Both coefficients are nonzero. The derivative term after rescaling has a uniform positive coefficient. These interval forms have uniformly positive Neumann coercivity: a sequence of unit L^2 functions with energy tending to zero would converge to a nonzero constant, while a limiting quadratic-square potential would force that constant to vanish. Thus their source responses are $O(R^{-3})$. Choosing K large enough places all local roots inside the interval, and the remaining local reciprocal-potential integral is also $O(R^{-3})$.

On the rootless side, bounded coordinate Jacobians in (98) give

$$\int_{\text{fold neighborhood}} k_c^{-1} ds \leq C \int_{\mathbb{R}} (|t| + x^2)^{-2} dx \leq C|t|^{-3/2}.$$

Every other root belongs to the uniform ordinary-root cover. The global lower bound $D_M \geq ca_R$, its nonzero slope, and (107) bound each such contribution by $Ca_R^{-1/4}$, even if that root is stationary

in c . The remaining positive-potential pieces contribute a constant. Allowing independent endpoint traces enlarges the variational class, so summing these interval responses bounds the full response from above. The number of intervals is uniformly bounded. This proves [equations \(104\) to \(106\)](#).

To finish the moment estimate, a finite sum of nonnegative responses raised to a fixed $q > 0$ is bounded by a fixed multiple of the sum of their q th powers. The bounded density and $|dc/dt| \asymp 1$ reduce the root-side contribution to

$$Ca_R^{-q/4} \int_R^{r^*} r^{e-1} dr, \quad e = 2 - \frac{3q}{2} + \frac{\alpha q}{4} = \begin{cases} 2 - 3q/2, & q \leq 2/3, \\ (8 - 5q)/(q + 4), & q > 2/3. \end{cases} \quad (108)$$

The ordinary-root contribution is $O(a_R^{-q/4})$; the central window contributes $O(R^{2-3q})$. The rootless integral is bounded when $q < 2/3$, logarithmic when $q = 2/3$, and $O(R^{2-3q})$ when $q > 2/3$.

For $q < 8/5$, $e > 0$ and $a_R \asymp M$, so the root integral is $O(M^{-q/4})$. The central power satisfies $R^{2-3q}/a_R^{-q/4} \asymp R^e \rightarrow 0$; any bounded or logarithmic rootless term is smaller as well. At $q = 8/5$, $\alpha = 1$ and $e = 0$, and the root integral is

$$O(a_R^{-2/5} \log(1/R)) = O(M^{-2/5} [\log(1/M)]^{7/5}).$$

The central and rootless terms have one fewer logarithmic factor. For $q > 8/5$, $e < 0$ and $a_R^{-q/4} R^e \asymp R^{2-3q} = M^{(2-3q)/7}$; the regular contribution is smaller. These are the upper bounds in [theorem 5.2](#), completing its proof.

5.4 Consequences for the physical transport problem

Let $\mathcal{G}_q^g(M)$ be the corresponding optimal full-bulk moment over physical predetermined fields, with the fixed geometry, positive bulk diffusivity, constant affinity, and nonzero V assumed in [section 2](#). The uniform bounds [\(97\)](#) verify [\(2\)](#). All three scales in [\(96\)](#) dominate $[1 + \log(1/M)]^q$. Therefore [theorem 2.4](#), with no observation, gives the stronger comparison

$$\mathcal{G}_q^g(M) \sim \chi^q \mathcal{P}_q^g(M). \quad (109)$$

In particular the same three moment orders hold for finite transverse bulk diffusion. The exact ratio follows by adding a vanishing uniform background to a scalar near-minimizer at the same total budget, as in [\(32\)](#); it does not require that the explicit graded trial be a sharp optimizer. Directly on that trial, $\min D_M \geq ca_R$ and $\log(1/a_R) = O_q(\log(1/M))$, so [theorem 2.3](#) bounds the bulk correction by a lower-order logarithm. No bounded correction or correction limit is asserted for a general family.

Only one positively sampled fold is used in the lower bounds; the upper design protects every possible fold site. Thus asymmetry, unequal root slopes, and additional regular roots change comparison constants without changing the threshold. In contrast, vanishing or singular sampling density at a fold, higher spatial degeneracy, failure of $g_c \neq 0$, or a family approaching an identically zero profile falls outside the proof. A law supported only on the rootless side need not have the moving-root contribution. Mobility caps, prescribed positive backgrounds, fixed spatial resolution, and uncertainty in the fold sites are also different design problems. The result uses the known fold geometry and an unrestricted integral budget; it does not claim generic sharp coefficients or a unique limiting design.

6 Exact observation and optimal local placement

Knowing the offset changes the design problem: mobility can now be placed around the observed kinetic zeros. We first solve the local quadratic problem over all integrable coefficients, then use explicit policies to justify averaging over offsets. The local model is an energy-dual problem on the line, not a stationary transport process on an infinite wall. Saturated-gradient certificates are an established method in reinforcement design [8]. Constant optimal gradients also occur in fixed-volume cooling-fin design [1]. The profile and constants needed here are derived directly for quadratic killing and a uniform source.

6.1 The quadratic placement problem and its global certificate

For $a, m > 0$, define

$$\mathcal{J}_a^{\text{quad}}(D) = \sup_{v \in C_c^\infty(\mathbb{R})} \left\{ 2 \int_{\mathbb{R}} v \, dx - \int_{\mathbb{R}} [D|v'|^2 + ax^2v^2] \, dx \right\}, \quad D \geq 0, \quad \int_{\mathbb{R}} D \, dx = m. \quad (110)$$

As in (9), an infinite response is allowed, and no operator realization for an arbitrary L^1 coefficient is presumed.

Theorem 6.1 (Exact quadratic placement). *Set*

$$R = \left(\frac{80m}{3a} \right)^{1/5}, \quad y = \frac{|x|}{R}, \quad C_{\text{pl}} = 12 \left(\frac{3}{80} \right)^{1/5}. \quad (111)$$

The unique minimizer of (110), up to equality almost everywhere, is

$$D_*(x) = \frac{aR^4}{8} y(1-y)^2(1+2y)\mathbf{1}_{\{y < 1\}}. \quad (112)$$

Its source representative and optimal response are

$$h_*(x) = \begin{cases} (9-8y)/(aR^2), & y \leq 1, \\ 1/(ax^2), & y > 1, \end{cases} \quad (113)$$

$$\inf_D \mathcal{J}_a^{\text{quad}}(D) = \mathcal{J}_a^{\text{quad}}(D_*) = \frac{12}{aR} = C_{\text{pl}} a^{-4/5} m^{-1/5}. \quad (114)$$

The source representative is understood in the completion in the reaction-plus-derivative energy. In particular, the theorem does not regard the constant source as an $L^2(\mathbb{R})$ vector.

Proof. Write $p(y) = y - 3y^3 + 2y^4$. It is nonnegative on $[0, 1]$, and

$$\int_0^1 p(y) \, dy = \frac{3}{20}, \quad p'(y) + y^2(9-8y) = 1. \quad (115)$$

The first identity gives $\int D_* = 3aR^5/80 = m$. On $0 < x < R$, $D_*h'_* = -Rp(x/R)$, so the second proves $-(D_*h'_*)' + ax^2h_* = 1$. Reflection gives the same equation on $(-R, 0)$; outside $[-R, R]$ it reduces to $ax^2h_* = 1$. The flux vanishes at $-R, 0, R$. Thus neither the cusp at zero nor the change of slope at the support edges introduces a delta source. Direct integration yields

$$\int h_* = \frac{12}{aR}, \quad \int D_*|h'_*|^2 = \frac{12}{5aR}, \quad \int ax^2h_*^2 = \frac{48}{5aR}. \quad (116)$$

The inside and outside contributions to $\int h_*$ are $10/(aR)$ and $2/(aR)$, respectively.

For every compact smooth v , integration of the weak equation gives

$$2 \int v - \int [D_* |v'|^2 + ax^2 v^2] = \int h_* - \int [D_* |(v - h_*)'|^2 + ax^2 (v - h_*)^2] \leq \int h_*.$$

The field h_* belongs to the stated energy completion: cut off its $|x|^{-2}$ tail at distance $L \rightarrow \infty$, then mollify its finitely many corners. Tail source and reaction integrals tend to zero, and the derivative energy is supported in $[-R, R]$, where D_* is bounded. These approximations approach h_* in energy and in source integral, proving equality in (114) for D_* .

The bound that excludes every competing placement is

$$b_* := \|h'_*\|_\infty = \frac{8}{aR^3}, \quad |h'_*| = b_* \quad (0 < |x| < R), \quad |h'_*| < b_* \quad (|x| > R). \quad (117)$$

For an arbitrary nonnegative $D \in L^1$ of mass m , the same cutoffs and mollifications have derivatives bounded by $b_* + o(1)$ and converging almost everywhere to h'_* . Dominated convergence against $D \, dx$ therefore makes them legitimate tests and gives

$$\begin{aligned} \mathcal{J}_a^{\text{quad}}(D) &\geq 2 \int h_* - \int ax^2 h_*^2 - \int D |h'_*|^2 \\ &\geq 2 \int h_* - \int ax^2 h_*^2 - mb_*^2 = \frac{12}{aR}. \end{aligned} \quad (118)$$

The derivative contribution at the large cutoff is $O(L^{-3})$; mollification does not increase its Lipschitz bound. This justifies (118) even for unbounded or highly concentrated competitors.

If equality holds, the strict inequality in (117) forces $D = 0$ almost everywhere outside $[-R, R]$. Also h_* maximizes the extended quadratic functional for D : its value equals the supremum, and $h_* + t\phi$ is a legitimate approximated test for every $\phi \in C_c^\infty$ and real t . The first variation gives $-(Dh'_*)' + ax^2 h_* = 1$ in distributions. Hence Dh'_* has an absolutely continuous representative locally, with zero exterior flux. On each open half-support, h'_* is a nonzero constant; integration from R or $-R$ uniquely recovers (112). This proves uniqueness almost everywhere. As a check on the mass normalization, $\partial_m \inf \mathcal{J}_a^{\text{quad}} = -64/(a^2 R^6) = -b_*^2$. $\square$

The optimal coefficient vanishes linearly at the kinetic zero and quadratically at its outer edges. These zeros do not invalidate the certificate, which only uses smooth tests and zero flux. They do mean that positivity or transmission across the center must not be imposed silently as an extra constraint. A fixed positive mobility floor, a pointwise cap, or a fixed fabrication length changes the problem.

6.2 Separated defects on a compact wall

The next result uses actual local masses of arbitrary competing fields. It therefore does not assume beforehand that an optimal sequence concentrates near the defects.

Theorem 6.2 (Compact-wall placement and allocation). *Let k be continuous on the compact wall Γ , positive except at finitely many distinct points $s_1, \dots, s_n$, with $n \geq 1$, and suppose*

$$k(s_j + x) = a_j x^2 [1 + o(1)], \quad a_j > 0, \quad x \rightarrow 0. \quad (119)$$

Then, over all nonnegative $L^1(\Gamma)$ fields of mass M ,

$$\inf_D J_k(D) \sim C_{\text{pl}} \left(\sum_{j=1}^n a_j^{-2/3} \right)^{6/5} M^{-1/5}. \quad (120)$$

A matching leading allocation to the individual defects is

$$m_j = M \frac{a_j^{-2/3}}{\sum_i a_i^{-2/3}}. \quad (121)$$

The kinetic profile and the separated defect positions are fixed as $M \downarrow 0$. The theorem does not apply uniformly as defects merge.

Proof. Fix $0 < \eta < 1/2$. Choose disjoint fixed neighborhoods $U_j = \{s_j + x : |x| < r_j\}$ in which $a_{j,-}x^2 \leq k(s_j + x) \leq a_{j,+}x^2$, where $a_{j,\pm} = (1 \pm \eta)a_j$. For an arbitrary competitor let $b_j = \int_{U_j} D$, so $\sum_j b_j \leq M$. For $b_j > 0$, take the field (113) with parameters $(a_{j,+}, b_j)$ and multiply it by a fixed smooth cutoff equal to one on $|x| < r_j/2$ and supported in U_j . For all small M , uniformly over $0 < b_j \leq M$, the cutoff acts only on the tail $1/(a_{j,+}x^2)$. It changes the source-minus-reaction integral by $O_\eta(1)$ and introduces only bounded slopes, whereas the central slope diverges at least as $M^{-3/5}$. Mollifying these compact Lipschitz tests retains that slope bound. Potential ordering and the actual mass b_j in each neighborhood give

$$J_k(D) \geq \sum_{j=1}^n C_{\text{pl}} a_{j,+}^{-4/5} b_j^{-1/5} - C_\eta. \quad (122)$$

The tests have disjoint supports, so their source and energy terms add. If $b_j = 0$, use the same test with an auxiliary positive budget tending to zero: its mobility cost is zero and its source-minus-reaction value diverges. Thus (122) holds with the value $+\infty$ in this case.

For positive weights w_j , elementary minimization gives

$$\inf_{b_j > 0, \sum b_j \leq M} \sum_j w_j b_j^{-1/5} = M^{-1/5} \left(\sum_j w_j^{5/6} \right)^{6/5}. \quad (123)$$

The objective decreases in each mass, so the total mass is M ; the derivative condition gives $b_j \propto w_j^{5/6}$, and strict convexity proves the minimum. Substitution into (122) yields the unrestricted lower bound.

For the upper bound allocate the masses (121) and insert (112), with curvature $a_{j,-}$ and mass m_j , into each neighborhood; set $D = 0$ elsewhere. The supports $|x| < R_j$ are disjoint for small M and the budget is exactly M . On a support, denote the interior field in (113) by $h_{j,-}$. Its zero-flux equation and $k \geq a_{j,-}x^2$ imply, for every smooth test restricted to that support,

$$2 \int_{-R_j}^{R_j} v - \int_{-R_j}^{R_j} [D|v'|^2 + kv^2] \leq \int_{-R_j}^{R_j} h_{j,-} = \frac{10}{a_{j,-}R_j}. \quad (124)$$

This is direct completion of the square; no boundary trace of v is constrained. Outside the supports use $2v - kv^2 \leq 1/k$. In $U_j \setminus [-R_j, R_j]$, the reciprocal integral is at most $2/(a_{j,-}R_j) + O_\eta(1)$, and the remaining fixed region contributes a bounded amount. Summing proves the upper counterpart of (122). Divide by $M^{-1/5}$, first send $M \rightarrow 0$, then $\eta \rightarrow 0$ to prove (120). The bounded errors at fixed η do not imply an additive bounded error after subtracting the final leading term with the exact a_j . $\square$

For comparison, the potential-ordering and interval-bracketing proof of [proposition 3.2](#) uses only the local quadratic asymptotic and a positive lower bound away from the zeros. It therefore applies to the present continuous rates and gives $J_k(M/P) \sim C_0 P^{1/4} \sum_j a_j^{-3/4} M^{-1/4}$. Placement changes the divergence exponent from $1/4$ to $1/5$ for a fixed set of observed quadratic defects.

Corollary 6.3 (Physical localized trials and their bulk correction). *Under the physical assumptions of [section 2](#) with $V \neq 0$, the infimum of D_{flow} over physical budget- M fields has leading value χ times [\(120\)](#). For the fixed- η localized trials used in its proof, the minimal closed derivative form is a valid realization and*

$$\|kh - 1\|_{L^2(\Gamma)} = O_\eta(M^{1/10}), \quad \mathcal{R}(D, k) \longrightarrow \mathcal{R}_0, \quad (125)$$

where $\mathcal{R}_0$ is defined by [\(55\)](#). These are fixed-kinetic-profile statements, not bounds uniform in an offset approaching a fold.

Proof. The leading full-bulk optimum follows immediately from [theorem 2.4](#) for a single fixed realization and [\(120\)](#); its positive-background mixture preserves the budget and asymptotic value. For the stronger assertion about the original degenerate trials, closability can be checked on every compact subinterval where $D > 0$. If smooth $v_l \rightarrow 0$ in L^2 and $\sqrt{D}v'_l$ is Cauchy in L^2 , its limit vanishes on each such subinterval by distributional differentiation. It is zero where $D = 0$, and the finitely many boundary points have measure zero. Thus the derivative preform is closable.

The endpoints of each half-support have zero derivative capacity: a transition of height one at an outer quadratic zero over width ϵ costs $O(\epsilon)$; near the central linear zero, a transition with derivative proportional to $1/x$ on $\epsilon^2 < x < \epsilon$ costs $O(1/|\log \epsilon|)$. The corresponding L^2 costs tend to zero. Smooth approximations to these transitions first localize bounded form functions to individual components: the cutoff derivative cost is bounded by the squared supremum of the function times the transition cost, while its remaining energy on the shrinking transition intervals tends to zero. Truncations $T_N(v) = \max(-N, \min(v, N))$ converge to v in the form norm as $N \rightarrow \infty$, extending this conclusion to all form-domain elements. Thus the minimal closure permits independent restrictions to the half-supports and exterior, with no endpoint matching condition; finite endpoint traces need not exist. This domain fact is used only for the explicit coefficients, not for arbitrary competing weights.

The reaction-plus-derivative form is coercive for each fixed $M > 0$. Indeed, away from the centers k is bounded below on the relevant closed sets. Near a center, anchor a test in an interval inside the half-support where D and k are positive. Weighted Cauchy–Schwarz and $D(x) \asymp_M |x|$ give $|v(x)|^2 \leq C_M \mathbf{a}_{k,D}[v](1 + |\log |x||)$; integrating this bound controls its remaining L^2 norm. Thus its inverse h exists in L^2 and in the minimal form domain. Positivity and potential comparison on each support give $0 \leq h \leq h_{j,-}$: subtract the two weak equations and test with the positive part of $h - h_{j,-}$, using the zero-capacity endpoints. Outside the supports $h = 1/k$ almost everywhere. Since

$$\max_{0 \leq y \leq 1} y^2(9 - 8y) = \frac{27}{16}, \quad 0 \leq kh \leq \frac{a_{j,+}}{a_{j,-}} \frac{27}{16} \quad \text{on support } j,$$

$kh - 1$ is bounded and supported on a set of length $O_\eta(M^{1/5})$. This proves the first assertion in [\(125\)](#). No transmission rule for an initial point mass at a kinetic zero is needed for this stationary, absolutely continuous formulation.

In [\(21\)](#), the load $\mathcal{L}_D$ consequently converges in the mean-zero bulk energy dual norm to $\mathcal{L}_0(f) = \int_\Omega (u - V)f - KV \int_\Gamma f_\Gamma$. Dropping the nonnegative Schur penalty gives $\limsup \mathcal{R} \leq \mathcal{R}_0$. For any fixed smooth bulk test f , choosing $v = f_\Gamma$ in that penalty gives $0 \leq \mathcal{S}_D(f_\Gamma) \leq M \|\partial_s f_\Gamma\|_\infty^2 \rightarrow 0$. Insert such tests in the supremum and use their density in the mean-zero bulk energy space to get

$\liminf \mathcal{R} \geq \mathcal{R}_0$. This proves convergence without assuming a bounded tangential derivative for the bulk maximizer. $\square$

6.3 The exact-observation mean and a measurable recovery policy

Return to the cosine ensemble (36) and define

$$\mathcal{O}(M) = F_1(M; c) = \inf_{c \rightarrow D_c \text{ Borel}, D_c \geq 0, \int D_c = M} \frac{1}{4} \int_{-2}^2 J_c(D_c) dc. \quad (126)$$

The budget is the same for every observed offset. This definition is an infimum over policies, rather than an unproved interchange of expectation and pointwise minimization.

Theorem 6.4 (Value of exact kinetic observation). *As $M \downarrow 0$,*

$$\mathcal{O}(M) \sim K_{\text{obs}} M^{-1/5}, \quad K_{\text{obs}} = \frac{2^{6/5} C_{\text{pl}}}{4} \text{B}\left(\frac{1}{2}, \frac{1}{5}\right). \quad (127)$$

The full-bulk optimum is $G_1(M; c) \sim \chi K_{\text{obs}} M^{-1/5}$ in the convention of [theorem 2.4](#). Compared with the predetermined mean,

$$\frac{\mathcal{O}(M)}{\mathcal{P}_1(M)} \sim \frac{K_{\text{obs}}}{K_1} M^{1/20} = 1.21856579 \dots M^{1/20}, \quad (128)$$

where $K_{\text{obs}} = 22.40462823 \dots$ and $K_1 = 18.38606365 \dots$. The same ratio holds for the two full-bulk mean optima.

We give the policy construction and its uniform estimates before completing the proof. Let $t = 1 - |c|$ and fix a sufficiently large number L . All constants below may depend on this fixed L , but not on M or c . Take M small enough that $LM^{2/7} < 1/2$. There is an explicit Borel policy $D_{M,c}$ with exact budget M such that

$$J_c(D_{M,c}) \leq C \begin{cases} M^{-1/5} t^{-4/5}, & t \geq LM^{2/7}, \\ M^{-3/7}, & |t| \leq LM^{2/7}, \\ |t|^{-3/2}, & t \leq -LM^{2/7}. \end{cases} \quad (129)$$

At each fixed $|c| < 1$ it also satisfies the sharp limit

$$M^{1/5} J_c(D_{M,c}) \longrightarrow 2^{6/5} C_{\text{pl}} (1 - c^2)^{-4/5}. \quad (130)$$

Separated roots. For $t \geq LM^{2/7}$ put $a = 1 - c^2$, let r_1, r_2 be the two roots, and set

$$z = \frac{M}{a^{7/2}}, \quad \eta = z^{1/10}, \quad b = a(1 - \eta), \quad \delta = \frac{\eta\sqrt{a}}{2}, \quad R = \left(\frac{40M}{3b}\right)^{1/5}. \quad (131)$$

Choosing L large makes $\eta \leq 1/2$ and $R \leq \delta/2$ throughout this regime: $z \leq L^{-7/2}$, $R/\delta = 2(40/3)^{1/5} z^{1/10} (1 - \eta)^{-1/5}$. The δ -neighborhoods of the roots are disjoint. Taylor's formula with $|\cos''| \leq 1$ gives

$$|\cos(r_j + x) - \cos r_j| \geq \sqrt{a}|x| - x^2/2, \quad k_c(r_j + x) \geq bx^2 \quad (|x| \leq \delta).$$

Place (112) with curvature b and mass $M/2$ on each root, and set mobility to zero elsewhere. The two masses add exactly to M .

For completeness, the reciprocal-potential estimate used outside these supports is

$$\int_{\Gamma \setminus \cup_j \{|s-r_j| < R\}} \frac{ds}{k_c(s)} \leq \frac{C}{aR} \quad (0 < R \leq c_1 \sqrt{a}), \quad (132)$$

with a fixed sufficiently small c_1 . Near each root, within a fixed small multiple of $\sqrt{a}$, Taylor's bound gives $k_c \geq ca|s-r_j|^2$, whose truncated reciprocal integral is $O((aR)^{-1})$. When c is near a fold, write x for arclength from that fold and $z_0 = 2\sin(x/2)$. Then $k_c = (z_0^2/2 - t)^2$, with bounded metric factors on a fixed fold neighborhood. Outside the two root neighborhoods, rescaling $z_0 = \sqrt{2t}y$ bounds the remaining reciprocal integral by $Ct^{-3/2} \int_{\text{dist}(y, \{-1,1\}) > c_2} (y^2 - 1)^{-2} dy = O(a^{-3/2})$. The part away from this fold is bounded. For offsets in a compact subset of $(-1, 1)$ the same estimate follows from a fixed positive potential away from the root neighborhoods. Finally $a^{-3/2} \leq C/(aR)$, proving (132). The support bound (124) and this estimate prove the first line of (129).

At fixed $|c| < 1$, $\eta \rightarrow 0$, $R/\delta \rightarrow 0$, and $b/a \rightarrow 1$. Within each δ -neighborhood the sharper exterior estimate is $2/(bR)$ per root, while the rest is $O_c(\delta^{-1}) = o(M^{-1/5})$, since $\delta \asymp_c M^{1/10}$. Together with the interior value $10/(bR)$ this gives the sharp upper bound in (130); the compact-wall lower theorem gives its matching lower bound.

Fold layer. For $|t| \leq LM^{2/7}$, use the fold at $s_0 = 0$ when $c < 0$, and at $s_0 = \pi$ when $c > 0$. Put $r = M^{1/7}$ and choose a fixed B with $B^2/2 > L + 2$. The policy is

$$D_{M,c}(s) = \frac{M}{2Br} \mathbf{1}_{\{|s-s_0| < Br\}}, \quad (133)$$

where the distance is the local periodic arclength. This has mass M . With $x = (s - s_0)/r$ and $\mu = t/r^2$, the potential on this fixed interval is $r^4 V_{r,\mu}(x)$, where

$$V_{r,\mu}(x) = \left(\frac{1 - \cos(rx)}{r^2} - \mu \right)^2 \longrightarrow (x^2/2 - \mu)^2$$

uniformly on $[-B, B] \times [-L, L]$. Since $D = r^6/(2B)$ on the patch, the diffusion operator has the same factor r^4 . For an unamplified rescaled test $v(s) = \phi(x)$, the integrated form and source are

$$r^5 \int_{-B}^B \left(\frac{|\phi'|^2}{2B} + V_{r,\mu} \phi^2 \right) dx, \quad r \int_{-B}^B \phi dx,$$

respectively. The free-endpoint form on $[-B, B]$ is uniformly coercive: Poincaré controls the non-constant part, and the integral of its potential is uniformly positive because no limiting polynomial $(x^2/2 - \mu)^2$ vanishes identically. A contradiction sequence with unit L^2 norm and energy tending to zero would converge to a nonzero constant with zero potential integral. Its source response is therefore bounded uniformly. Since the response scale is $r^2/r^5 = r^{-3}$, restricting any smooth wall test to the patch bounds its contribution by Cr^{-3} .

Outside the patch $D = 0$. Its boundary has a fixed scaled margin beyond all roots. More explicitly, for $Br \leq |s - s_0| \leq x_0$ and small fixed x_0 , $1 - \cos(s - s_0) - t \geq c|s - s_0|^2$ after enlarging B if necessary. Thus the reciprocal integral outside the patch is at most $C \int_{Br}^{x_0} x^{-4} dx + C \leq Cr^{-3}$. This proves the middle line of (129) for every original smooth wall test.

Rootless offsets and measurability. For $t \leq -LM^{2/7}$ use the uniform field $M/(2\pi)$. Dropping the nonnegative derivative energy gives $J_c(D) \leq \int_{\Gamma} k_c^{-1}$. Near the closest fold, $k_c(s) \geq c(|t| + |s - s_0|^2)^2$; rescaling by $|t|^{1/2}$ gives the last bound in (129). Offsets away from both folds have a bounded reciprocal integral, as required by that same bound for $-1 \leq t < 0$. The policy just constructed is Borel into $L^1(\Gamma)$: $r_1(c) = \arccos(-c)$ and $r_2(c) = 2\pi - r_1(c)$ vary continuously in the separated regime, as do its positive profile widths and amplitudes; translations and dilations of these compact continuous profiles are continuous in L^1 . The two fold patches and uniform branch are Borel, and the regime boundaries are Borel sets. Assign either adjoining prescription at a threshold, and the fold patch at $c = \pm 1$. Thus the policy is defined with exactly the required budget at every offset, including the null exceptional points.

Proof of theorem 6.4. The estimates just proved give the integrable, M -independent bound

$$M^{1/5} J_c(D_{M,c}) \leq C|1 - |c||^{-4/5}, \quad c \neq \pm 1. \quad (134)$$

For the middle regime this follows from $M^{1/5-3/7} = M^{-8/35}$ and $|t| \leq LM^{2/7}$; for the rootless regime it follows from $M^{1/5}|t|^{-7/10} \leq C$ when $|t| \geq LM^{2/7}$. At fixed $|c| > 1$, the scaled response tends to zero. Dominated convergence, (130), and the density $1/4$ therefore give

$$\limsup_{M \rightarrow 0} M^{1/5} \mathcal{O}(M) \leq \frac{2^{6/5} C_{\text{pl}}}{4} \int_{-1}^1 (1 - c^2)^{-4/5} dc = K_{\text{obs}}.$$

For the converse take any sequence $M_l \downarrow 0$ and Borel policies whose expected values are at most $\mathcal{O}(M_l) + 1$. Such policies exist by the definition of the finite infimum; no pointwise minimizing selection is used. At each fixed $|c| < 1$, theorem 6.2, applied to its two separated zeros of curvature $a = 1 - c^2$, bounds every member of this sequence and implies

$$\liminf_{l \rightarrow \infty} M_l^{1/5} J_c(D_{M_l,c}) \geq 2^{6/5} C_{\text{pl}} (1 - c^2)^{-4/5}.$$

Fatou's lemma applied to these nonnegative, measurable responses proves the matching lower bound after integration over $(-1, 1)$. This argument avoids any measurability assertion for a pointwise infimum over the uncountable coefficient class. The full-bulk statement follows from theorem 2.4, and the ratio follows from theorem 4.1 at $q = 1$. $\square$

The fold layer has width $O(M^{2/7})$ and contributes only $O(M^{-1/7})$ to the expected trial response. It is needed for a valid global policy, but is smaller than the mean's leading $M^{-1/5}$ term. Exact observation changes the exponent because each realization can receive mobility at its actual defects; the total budget is not redistributed among realizations. The ratio (128) tends to zero slowly and does not imply a specified improvement at any particular finite experimental budget. All coefficients in this section use the dimensionless normalization (6); the dimensional response factor must be restored before multiplication by the physical χ .

7 Finite measurement precision and the exact crossover

Suppose the designer observes the offset bin containing c , rather than its exact value. For an integer $N \geq 1$, let $\Delta = 4/N$ and partition $[-2, 2]$ at $-2 + j\Delta$, $j = 0, \dots, N$. The bins are half-open, except at the final endpoint. Write $\mathcal{I}_{\Delta}$ for this partition and define

$$\mathcal{Q}(M, \Delta) = \inf_{\substack{D_I \in L^1(\Gamma), D_I \geq 0 \\ \int_{\Gamma} D_I = M \text{ for every } I}} \frac{1}{4} \sum_{I \in \mathcal{I}_{\Delta}} \int_I J_c(D_I) dc. \quad (135)$$

The budget is available separately for every observed bin; it is not shared across bins. There are finitely many decisions at each M, Δ , so this is a measurable policy in the sense of [section 2.5](#). Its value is the sum of the conditional infima, multiplied by $\Delta/4$: choose an arbitrarily close competitor in each of the finitely many bins to obtain the reverse inequality. This does not require existence of a circle optimizer.

The next theorem includes the intermediate regime, in which the measurement error and the optimal placement width have comparable size. Its coefficient is expressed through a fully specified local optimization problem, rather than an elementary formula.

Theorem 7.1 (Measurement precision and mobility budget). *For the cosine ensemble, jointly as $M \downarrow 0$ and $\Delta = 4/N \downarrow 0$,*

$$\mathcal{Q}(M, \Delta) \asymp M^{-1/5} + M^{-1/4} \Delta^{1/4}, \quad (136)$$

with constants independent of the relative rates of the two limits. Let $\mathcal{F}_{\text{ctr}}$ be the local value in [\(141\)](#) below and put

$$\mathcal{H}(\tau) = \frac{2^{6/5}}{4} \int_{-1}^1 (1 - c^2)^{-4/5} \mathcal{F}_{\text{ctr}} \left(2^{1/5} \tau (1 - c^2)^{-3/10} \right) dc, \quad \tau \geq 0. \quad (137)$$

This integral is finite and defines a continuous function. For every finite $\tau \geq 0$,

$$\frac{\Delta}{M^{1/5}} \longrightarrow \tau \implies M^{1/5} \mathcal{Q}(M, \Delta) \longrightarrow \mathcal{H}(\tau). \quad (138)$$

In particular, $\mathcal{H}(0) = K_{\text{obs}}$. In the separate coarse limit $\Delta \downarrow 0$, $\Delta/M^{1/5} \rightarrow \infty$,

$$\mathcal{Q}(M, \Delta) \sim K_{\text{coarse}} M^{-1/4} \Delta^{1/4}, \quad K_{\text{coarse}} = 2^{-3/4} C_0 \mathcal{B}(1/2, 1/8). \quad (139)$$

Also $\mathcal{H}(\tau) \sim K_{\text{coarse}} \tau^{1/4}$ as $\tau \rightarrow \infty$. These limits do not depend on whether a fold lies at a bin edge or inside a bin.

7.1 A quadratic defect with an uncertain center

For a finite nonnegative Radon measure ν on $\mathbb{R}$, define

$$\mathcal{I}_z(\nu) = \sup_{f \in C_c^\infty(\mathbb{R})} \left\{ 2 \int_{\mathbb{R}} f dx - \int_{\mathbb{R}} |f'|^2 d\nu - \int_{\mathbb{R}} (x - z)^2 f^2 dx \right\}, \quad (140)$$

$$\mathcal{F}_{\text{ctr}}(\eta) = \inf_{\substack{d \geq 0, \\ d \in L^1(\mathbb{R}) \\ \int_{\mathbb{R}} d = 1}} \int_{-1/2}^{1/2} \mathcal{I}_{\eta v}(d dx) dv, \quad \eta \geq 0. \quad (141)$$

Here v in the second line is a scalar parameter. Thus $\eta > 0$ means a center uniformly distributed over an interval of width η , whereas $\eta = 0$ means a known center. Measures will only be used for compactness; the admissible mobility fields in [\(141\)](#) are densities. The response is an energy-dual quantity on an infinite line, not a stationary transport coefficient there.

For a physical local coordinate with potential $a(x - z)^2$, allocated mass m , and center uncertainty width w , set $\ell = (m/a)^{1/5}$. The substitutions $x = \ell y$, $D(x) = a \ell^4 d(y)$, and $h(x) = (a \ell^2)^{-1} f(y)$ give the exact conditional optimum

$$a^{-4/5} m^{-1/5} \mathcal{F}_{\text{ctr}}(w/(m/a)^{1/5}). \quad (142)$$

Translation and reflection preserve the mass constraint. Convexity of the response permits an even d in [\(141\)](#).

We first give the compactness fact needed to pass from this local problem to the bin-dependent optimum. Its positive tail is a proof device that can be added with arbitrarily small mass.

Lemma 7.2 (Quadratic response under expanding-interval limits). *Suppose $d \in L^1(\mathbb{R})$ and $d(x) \geq b(1 + |x|)^{-\alpha}$ for some $b > 0$ and $1 < \alpha < 2$. Then $z \mapsto \mathcal{T}_z(d dx)$ is finite and continuous. Let $L_n^-, L_n^+ \rightarrow \infty$. On $(-L_n^-, L_n^+)$ take the free-endpoint response with derivative energy $\int d|f'|^2$, source $\int w_n f$, and reaction energy $\int w_n(x - z_n)^2 f^2$. If $z_n \rightarrow z$, the positive weights w_n are uniformly bounded above and below and converge locally uniformly to one, then the interval response tends to $\mathcal{T}_z(d dx)$. The conclusion is uniform over compact sets of center values, in the sequential sense just stated, and is unchanged if the derivative coefficient is multiplied by a scalar tending to one.*

Proof. The proof parallels lemma 4.5, with a quadratic tail in place of a quartic one; the slower source tail must be retained. On a fixed core, a lower bound on d and a subinterval separated from all the possible centers control the H^1 norm by the energy. Outside that core $(x - z_n)^2 \geq cx^2$. Consequently, writing ϵ_n for the interval energy,

$$\left| \int w_n f \right| \leq C \epsilon_n[f]^{1/2}, \quad \int_{|x|>L} |w_n f| dx \leq CL^{-1/2} \epsilon_n[f]^{1/2}. \quad (143)$$

The integrals are restricted to the interval where necessary. Maximizing sequences can therefore be chosen with uniformly bounded energy. They have locally weak H^1 limits and locally weak weighted derivatives. Distributional differentiation identifies the limiting derivative, and lower semicontinuity controls its energy. The source integrals converge by (143).

To verify that the limit belongs to the whole-line smooth-test completion, apply the one-dimensional Sobolev estimate on $[x - 1, x + 1]$. Its local function and derivative squared norms are bounded respectively by $C|x|^{-2}\epsilon[f]$ and $C|x|^\alpha\epsilon[f]$. Thus

$$|f(x)|^2 \leq C(|x|^{-2} + |x|^{\alpha/2-1})\epsilon[f] \quad (144)$$

for large $|x|$. In particular the limit is bounded. A cutoff between R and $2R$ adds derivative energy at most $CR^{-2}\|f\|_\infty^2 \int_{R<|x|<2R} d$, which vanishes; the original energy tails vanish as well. On compact intervals, approximate the derivative in $L^2(d dx)$ by smooth functions, correct its integral with a fixed smooth bump when both traces are prescribed, and integrate. The compact lower bound on d makes this approximation uniform in the function, so source and reaction also converge. This proves density in the required energy space, including for unbounded d . The limiting value is therefore no greater than the whole-line response. Conversely, any compact smooth test is eventually admissible and its weighted integrals converge, which proves the opposite inequality.

For continuity on the whole line, the same compact-core estimate gives $\int (1 + x^2) f^2 \leq C \epsilon_z[f]$ for centers in a fixed compact set. Since $|(x - z)^2 - (x - z')^2| \leq C|z - z'|(1 + x^2)$ there, nearby forms bound one another with factors tending to one. The source is bounded in those energy norms by (143). Their response values are consequently continuous. The compactness proof also applies along converging center sequences and to scalar changes of the derivative coefficient, proving the final assertions. $\square$

Proposition 7.3 (Local value and its endpoint limits). *For every $\eta \geq 0$, the infimum in (141) is attained by a nonnegative L^1 density of mass one; an even minimizer can be chosen. The value is continuous on $[0, \infty)$ and satisfies*

$$\max\{C_{\text{pl}}, C_0\eta^{1/4}\} \leq \mathcal{F}_{\text{ctr}}(\eta) \leq C(1 + \eta^{1/4}), \quad \mathcal{F}_{\text{ctr}}(0) = C_{\text{pl}}. \quad (145)$$

Moreover,

$$\mathcal{F}_{\text{ctr}}(\eta) \sim C_0\eta^{1/4} \quad (\eta \rightarrow \infty). \quad (146)$$

No uniqueness or monotonicity in the interval width is needed for these assertions or for theorem 7.1.

Proof. Allowing the center to be known can only reduce the objective. The translated quadratic placement theorem therefore gives $\mathcal{T}_z(d\,dx) \geq C_{\text{pl}}$ for every unit-mass d and every z , including an infinite response. This proves the first lower bound and equality at zero.

For the other lower bound fix $\psi \in C_c^\infty(\mathbb{R})$ and write $I_\psi = \int \psi$, $T_\psi = \int |\psi'|^2$, $U_\psi = \int x^2 \psi^2$. For $\eta > 0$ use the translated tests

$$h_z(x) = \eta^{1/2} \psi((x - z)/\eta^{-1/4}), \quad |z| \leq \eta/2.$$

Their source minus reaction is $\eta^{1/4}(2I_\psi - U_\psi)$, independently of z . At every x , the averaged squared derivative is bounded by

$$\frac{1}{\eta} \int_{-\eta/2}^{\eta/2} |h'_z(x)|^2 \, dz \leq \eta^{1/4} T_\psi.$$

Integrating this bound against any unit-mass mobility gives $\mathcal{F}_{\text{ctr}}(\eta) \geq \eta^{1/4}(2I_\psi - U_\psi - T_\psi)$. Take the oscillator variational supremum to obtain $C_0 \eta^{1/4}$. This lower certificate allows arbitrary mobility spikes and zero sets.

For an upper bound put constant mobility $1/(\eta + 2)$ on $[-\eta/2 - 1, \eta/2 + 1]$ and zero outside. Neumann bracketing and reciprocal-potential integration outside this interval give a bound $C(1 + \eta^{1/4})$. When $\eta \rightarrow \infty$, the harmonic length $(\eta + 2)^{-1/4}$ is much smaller than the unit padding. The natural-endpoint oscillator limit is uniform over all center positions, so the interior contribution is $C_0(\eta + 2)^{1/4}[1 + o(1)]$. The exterior contribution is at most two. This proves (146).

For compactness, the singular-mass removal proof in lemma 4.4 applies to (140) without change: derivative flattening near a null support changes the source and the locally bounded reaction by $o(1)$. Hence

$$\mathcal{T}_z(d\,dx + \nu_s) = \mathcal{T}_z(d\,dx). \quad (147)$$

For a compact smooth test its value is continuous under vague convergence of finite measures and simultaneous convergence of z . A common countable set of compact smooth tests, on a countable exhaustion of the line, gives measurability of the supremum. That supremum is jointly lower semicontinuous.

Take a minimizing sequence of unit-mass densities for fixed η . It has a vaguely converging subsequence with limit ν of mass at most one. Fatou's lemma, after the fixed parameterization $z = \eta v$, bounds the objective at ν by the liminf. Remove its singular part using (147), and add a nonnegative density to replace any missing mass. Increasing the derivative coefficient cannot increase the response. The resulting unit-mass density therefore has value no greater than the infimum and is a minimizer. This argument makes no assumption that the original minimizing sequence was tight, or that its weak limit was absolutely continuous. Convex reflection averaging gives an even minimizer. The same argument with $\eta_n \rightarrow \eta$ proves lower semicontinuity of $\mathcal{F}_{\text{ctr}}$.

For the upper semicontinuity, choose a unit-mass density d with objective within ε of the value at η . Let ρ_α be a positive unit-mass multiple of $(1 + |x|)^{-\alpha}$, $1 < \alpha < 2$, and set

$$d_\varepsilon = \frac{d + \varepsilon \rho_\alpha}{1 + \varepsilon}. \quad (148)$$

It has mass one and a positive integrable tail. Form ordering gives $\mathcal{T}_z(d_\varepsilon\,dx) \leq (1 + \varepsilon)\mathcal{T}_z(d\,dx)$. By lemma 7.2, the response of this fixed regularization is continuous, uniformly for centers in a compact set. Its average is thus continuous in η . Use it as a competitor at η_n , and then let $\varepsilon \downarrow 0$. This proves upper semicontinuity and completes the proof. $\square$

7.2 Uniform localization of a conditional optimum

For a bin I with midpoint $c_I \in (-1, 1)$, set

$$\begin{aligned} a_I &= 1 - c_I^2, \quad r_I = \arccos(-c_I), \quad m = M/2, \quad \ell_I = (m/a_I)^{1/5}, \\ \eta_I &= \frac{\Delta}{\sqrt{a_I}\ell_I} = 2^{1/5} \frac{\Delta}{M^{1/5}} a_I^{-3/10}. \end{aligned} \quad (149)$$

Let $V(M, I)$ be the infimum of the conditional mean $\Delta^{-1} \int_I J_c(D) \, dc$ over all budget- M fields.

Lemma 7.4 (Conditional variational localization). *On every fixed compact $K \subset (-1, 1)$, uniformly for bins with $c_I \in K$ as $M, \Delta \downarrow 0$ and $\Delta/M^{1/5}$ remains bounded,*

$$\frac{V(M, I)}{2a_I^{-4/5}(M/2)^{-1/5}} - \mathcal{F}_{\text{ctr}}(\eta_I) \longrightarrow 0. \quad (150)$$

The lower bound allows every nonnegative L^1 field. The upper bound is realized by a field with exactly mass M selected using only the bin, and can be made uniform on K .

Proof. Reflection $s \mapsto -s$ preserves k_c for each c , not just after ensemble averaging. Convexity therefore allows replacement of any competitor by $[D(s) + D(-s)]/2$ without increasing its conditional cost. The two half-circles then have exactly mass $m = M/2$. This symmetry does not require that I be symmetric under $c \mapsto -c$.

For the lower bound consider any sequence of bins as in the statement, and choose a subsequence attaining the liminf of the normalized conditional values. By further subsequences, $c_I \rightarrow c_* \in K$ and $\eta_I \rightarrow \eta_*$. Take symmetric competitors whose normalized costs approach that liminf. Push their mass on $(0, \pi)$ to $x = (s - r_I)/\ell_I$, dividing it by m . These probability measures on expanding intervals have a vaguely converging subsequence with limit ν , of mass at most one. For $c = c_I + \Delta v$, $|v| \leq 1/2$, the scaled potential satisfies, locally uniformly in x, v ,

$$\frac{c + \cos(r_I + \ell_I x)}{\sqrt{a_I}\ell_I} = \eta_I v - x + O_K(\ell_I x^2).$$

Insert any compact smooth test at r_I , with amplitude $(a_I \ell_I^2)^{-1}$, and its reflected copy at $-r_I$. The supports are disjoint for small M . The scaled source and reaction converge to those in (140), and the derivative integral converges vaguely. Taking the supremum over compact tests gives, for each v ,

$$\liminf \frac{J_{c_I + \Delta v}(D_I)}{2a_I^{-1}\ell_I^{-1}} \geq \mathcal{T}_{\eta_* v}(\nu). \quad (151)$$

Fatou's lemma then gives the averaged lower bound $\int_{-1/2}^{1/2} \mathcal{T}_{\eta_* v}(\nu) \, dv \geq \mathcal{F}_{\text{ctr}}(\eta_*)$: remove singular mass and complete the remaining density to mass one as in [proposition 7.3](#). Continuity of the local value turns this sequential bound into the uniform lower assertion. No regularity, scale separation within the design, or absence of escaped mass has been imposed on the competitor.

For recovery fix a limit pair c_*, η_* and a unit-mass profile d_ϵ as in (148), whose averaged value at η_* is within an arbitrarily small error of $\mathcal{F}_{\text{ctr}}(\eta_*)$. In a fixed small neighborhood of r_I , use the exact coordinate

$$x = \frac{-\cos s - c_I}{\sqrt{a_I}\ell_I}, \quad ds = \ell_I w_I(x) \, dx, \quad w_I(x) = \frac{\sqrt{a_I}}{\sin s}. \quad (152)$$

The neighborhood stays inside $(0, \pi)$ uniformly on K . Its scaled endpoints tend to both infinities; w_I is uniformly bounded above and below and tends locally uniformly to one. For every c in the bin,

$$k_c(s) = a_I \ell_I^2 (x - \eta_I v)^2, \quad c = c_I + \Delta v,$$

exactly. Put the preliminary field

$$D_I(s) = a_I \ell_I^4 w_I(x) d_\varepsilon(x) \quad (153)$$

on this neighborhood, its reflection on the other root neighborhood, and zero elsewhere. Its mass on each neighborhood is $m \int w_I^2 d_\varepsilon dx = m[1 + o(1)]$ by dominated convergence. A single scalar normalization therefore makes its total mass exactly M and tends to one.

Under the response scaling $(a_I \ell_I)^{-1}$, the derivative energy in (153) is precisely $\int d_\varepsilon |f'|^2$; source and reaction have the weight w_I . Apply Neumann bracketing at the neighborhood endpoints and lemma 7.2. The exterior reciprocal-potential integral is bounded on K , because all possible roots lie strictly inside the two neighborhoods. Its contribution vanishes after multiplication by $a_I \ell_I$. Consequently this recovery has normalized conditional value tending to $\int_{-1/2}^{1/2} \mathcal{J}_{\eta_* v}(d_\varepsilon dx) dv$. Normalization changes the response by a factor tending to one, as the full forms are bounded by the minimum and maximum of one and that normalization factor. Let $\varepsilon \downarrow 0$.

These arguments prove the upper assertion along every converging sequence in the compact parameter set. Equivalently, one may choose a finite collection of regularized profiles: continuity of their averaged responses supplies a finite cover of the compact range of η_I . Assign the first qualifying profile to each bin and use (153). This gives uniform errors, a deterministic bin policy, and an exact per-bin budget. Together with the sequential lower bound it proves (150). $\square$

7.3 Global bounds, including bins crossing a fold

The regular-bin limit is insufficient by itself: curvatures vanish at $c = \pm 1$, and some bins contain both rootless and two-root realizations. We record a single family of admissible trials that controls all these bins uniformly.

Lemma 7.5 (Bin policies with uniform integrable bounds). *Put $W = \Delta + M^{2/7}$. There are fixed constants such that bins on the two-root side at distance $t = 1 - |c| \geq CW$ from the nearest fold admit a common-within-bin field satisfying*

$$J_c(D_I) \leq C \left[M^{-1/5} t^{-4/5} + M^{-1/4} \Delta^{1/4} t^{-7/8} \right] \quad (c \in I). \quad (154)$$

All bins intersecting a fixed multiple of the fold neighborhoods $|1 - |c|| \leq CW$ can be covered by two groups. Each group admits a single budget- M field whose integrated response over that group is at most

$$CM^{-1/4} W^{3/8}. \quad (155)$$

The integrated cost of the remaining rootless bins is at most $CW^{-1/2}$ and is absorbed in (155).

Proof. In a regular bin, t is comparable to its midpoint value. Each possible-root arc has length at most Cw , where $w = \Delta/\sqrt{t}$. Pad each arc by a fixed multiple of $\ell = (M/t)^{1/5}$ and put half the budget uniformly on each padded arc. The patches have length comparable to $w + \ell$, and their mobility e is comparable to $M/(w + \ell)$. Since $t \geq CW$, increasing the fixed C if necessary makes the patches disjoint and keeps them in quadratic neighborhoods:

$$\frac{w}{\sqrt{t}} = \frac{\Delta}{t} \ll 1, \quad \frac{\ell}{\sqrt{t}} = (M/t^{7/2})^{1/5} \ll 1.$$

The constants denoted by $\ll 1$ here are fixed sufficiently small constants, uniformly over the construction. The harmonic length $(e/t)^{1/4}$ is at most a constant times ℓ , so every possible root has a fixed positive margin in harmonic units. The interval oscillator bound and quadratic potential comparison give an interior response $Ce^{-1/4}t^{-3/4}$. Outside the patches, reciprocal-potential integration gives $C/(t\ell) + Ct^{-3/2}$, which is bounded by the same scale. Expanding $(w + \ell)^{1/4}$ proves (154).

Classify as fold bins every bin meeting the stated fold neighborhood. Since a bin has width $\Delta \leq W$, their union remains inside a possibly larger fixed multiple of that neighborhood, irrespective of grid alignment. Around the corresponding fold site s_j put constant mobility on an interval of radius $R = K_0\sqrt{W}$, with fixed K_0 large enough to contain every possible root with a fixed relative margin. Normalize it to mass M , so $e \asymp M/\sqrt{W}$, and put zero mobility outside. In the exact sine coordinate $z = 2\sin((s - s_j)/2)$ used in the proof of lemma 3.5, the local potential is $(z^2/2 - t)^2$, with uniformly comparable metric factors. The inequality $W \geq M^{2/7}$ ensures $W \geq ce^{1/3}$. The quartic inner layer and both tails from lemma 3.5 therefore give

$$\begin{aligned} \int_{|t| \leq CW} J_{\text{local}}(t) dt &\leq C \left[e^{-1/6} + e^{-1/4} \int_{e^{1/3}}^{CW} t^{-3/4} dt \right] \\ &\leq Ce^{-1/4} W^{1/4} \leq CM^{-1/4} W^{3/8}. \end{aligned}$$

An empty integral is omitted, and fixed comparison factors in the lower integration limit are harmless. The same bounds hold for the finite interval with free endpoints: each scaled root is separated from its endpoint by a fixed fraction of the interval radius. They follow from the interval bracketing and localization proof of lemma 3.5, not from an unrestricted large-parameter whole-line limit.

Outside the patch the reciprocal-potential response is at most $CW^{-3/2}$ per offset, hence $CW^{-1/2}$ after integration. This is absorbed above because $M^{1/4}W^{-7/8} \leq 1$. Finally, on the rootless side beyond these bins, $J_c(D) \leq \int_{\Gamma} k_c^{-1}$ for any field. The rootless reciprocal-potential bound integrates to $CW^{-1/2} + C$, also absorbed. All trial fields have exactly the prescribed mass, including those reused over an entire group of bins. $\square$

We can now prove the order theorem without imposing any scale condition on the relative precision. The lower bound $\mathcal{Q}(M, \Delta) \geq \mathcal{O}(M) \geq cM^{-1/5}$ follows from exact observation. For the coarse contribution retain bins in $[-1/2, 1/2]$ and assume $\Delta \geq LM^{1/5}$ with L a sufficiently large fixed number. In each retained bin the possible roots range over arcs of length comparable to Δ , with conditional densities at most C/Δ . Use compact nonnegative translated bumps at these roots, of width $b = (M/\Delta)^{1/4}$ and amplitude γb^{-2} . Their supports lie in uniformly quadratic neighborhoods, and the two root supports are disjoint. Averaging source minus reaction gives at least $c\gamma b^{-1} - C\gamma^2 b^{-1}$. At any spatial point the averaged squared derivative is at most $C\gamma^2/(\Delta b^5)$. Against an arbitrary total mass M its cost is at most $C\gamma^2 b^{-1}$. Choose a fixed sufficiently small $\gamma > 0$. This proves a conditional lower bound $cM^{-1/4}\Delta^{1/4}$ for every field in every retained bin. Their total probability is bounded below as $\Delta \rightarrow 0$. When $\Delta < LM^{1/5}$ this term is already controlled by the oracle lower bound. The maximum of the two bounds is comparable to their sum.

Integrating (154) gives the two terms in (136), since $4/5$ and $7/8$ are both less than one. The remaining terms satisfy

$$M^{-1/4}W^{3/8} \leq C(M^{-1/7} + M^{-1/4}\Delta^{3/8}) = o(M^{-1/5} + M^{-1/4}\Delta^{1/4}). \quad (156)$$

This proves the joint order law. For any fixed N the separate statement is $\mathcal{Q}(M, 4/N) \asymp M^{-1/4}$, with constants allowed to depend on N . A positive-length regular portion of one bin gives the same moving-root lower bound, and the predetermined mean design from theorem 4.1 supplies the upper bound. No small- Δ coefficient is asserted for that fixed partition.

7.4 Assembly of the exact crossover and sharp coarse limit

Suppose first that $\Delta/M^{1/5} \rightarrow \tau < \infty$. On a fixed compact subinterval of $(-1, 1)$, [lemma 7.4](#) and the finite-bin decomposition of (135) give Riemann sums with density

$$\frac{2^{6/5}}{4}(1-c^2)^{-4/5}\mathcal{F}_{\text{ctr}}\left(2^{1/5}\tau(1-c^2)^{-3/10}\right). \quad (157)$$

Uniformity in that lemma, and continuity of the local value, justify these sums even though the designs and number of bins both change. Discarding all other bins gives the global liminf; then enlarge the compact interval to $(-1, 1)$.

For the matching upper bound, use the recovery policies on the retained bins and the policies of [lemma 7.5](#) elsewhere. Any bins cut by the artificial ends of the retained interval have width at most Δ and can be assigned to the omitted set. The normalized regular-bin envelope is

$$Ct^{-4/5} + C(\Delta/M^{1/5})^{1/4}t^{-7/8},$$

whose integral over $0 < t < \delta$ is at most $C\delta^{1/5} + C\sup(\Delta/M^{1/5})^{1/4}\delta^{1/8}$. The normalized fold contribution tends to zero. Indeed $M^{1/5}M^{-1/7} = M^{2/35} \rightarrow 0$ and, for bounded $\Delta/M^{1/5}$, $M^{-1/20}\Delta^{3/8} = O(M^{1/40}) \rightarrow 0$. Send $M, \Delta \rightarrow 0$ first and then $\delta \downarrow 0$ to prove (138), including $\tau = 0$. The bound (145) also shows that the integrand in (137) is bounded on bounded τ sets by an integrable multiple of $(1-c^2)^{-4/5} + (1-c^2)^{-7/8}$. Dominated convergence proves continuity of $\mathcal{H}$ and

$$\mathcal{H}(0) = \frac{2^{6/5}C_{\text{pl}}}{4} \text{B}(1/2, 1/5) = K_{\text{obs}}.$$

The coarse limit requires its own uniform argument; letting $\tau \rightarrow \infty$ after a fixed- τ theorem would not prove a simultaneous limit. Retain again a compact regular offset set. In a bin with midpoint c_I , each root arc has length $w = \Delta/\sqrt{a_I}[1 + o(1)]$, uniformly on this set. Put $e = M/(2w)$ and $b = (e/a_I)^{1/4}$. The coarse hypothesis is equivalent here to $b/w \rightarrow 0$. Take harmonic tests centered at the two actual roots, with amplitude $(ea_I)^{-1/2}$, scale b , and a fixed compact smooth shape ψ . Set $z_0 = e^{-1/4}a_I^{-3/4}$. Their averaged source minus reaction is

$$2z_0(2I_\psi - U_\psi) + o(z_0).$$

The conditional root density on each arc is $(1 + o(1))/w$. The two neighborhoods are disjoint, so at every spatial point the sum of the two averaged derivative kernels is at most

$$[1 + o(1)]\frac{T_\psi}{w}e^{-5/4}a_I^{-3/4}.$$

Bin edges only truncate this nonnegative kernel. Multiplication by any competing mass $M = 2ew$ gives a derivative cost at most $[1 + o(1)]2z_0T_\psi$. First take the joint limit with ψ fixed, then take its oscillator variational supremum. This proves the unrestricted conditional lower bound $[1 - o(1)]2C_0z_0$. This proof does not presume an allocation by the competitor.

For the upper bound, divide the mass equally over the two root arcs, with padding p chosen so that $b \ll p \ll w$; for example $p = \sqrt{bw}$ works uniformly on the retained set. The padded constant mobility is $e[1 + o(1)]$. Neumann harmonic localization is uniform because every root is many harmonic lengths from each endpoint. The exterior reciprocal-potential contribution is $O((a_I p)^{-1}) = o(z_0)$. Curvature and the offset-to-root Jacobian vary by $o(1)$ on these shrinking arcs. Both bounds therefore give

$$V(M, I) = [1 + o(1)]2^{5/4}C_0a_I^{-7/8}M^{-1/4}\Delta^{1/4} \quad (158)$$

uniformly on every fixed regular offset set.

After dividing (154) by the coarse scale, the omitted regular-bin integrand is bounded by $C(M^{1/5}/\Delta)^{1/4}t^{-4/5} + Ct^{-7/8}$. In this regime $W \sim \Delta$, so the normalized fold term is $O(\Delta^{1/8}) \rightarrow 0$. Riemann sums in (158) thus extend to the full ensemble, with coefficient

$$\frac{2^{5/4}C_0}{4} \int_{-1}^1 (1-c^2)^{-7/8} dc = 2^{-3/4}C_0 B(1/2, 1/8).$$

This proves (139). Finally, applying (146) in (137) and using the same integrable bound gives $\mathcal{H}(\tau)/\tau^{1/4} \rightarrow K_{\text{coarse}}$. This last consistency statement follows from the local value; the simultaneous coarse limit has been proved separately.

7.5 Transport consequence and meaning of the resolution scale

For the full-bulk optimum $\mathcal{Q}_{\text{bulk}}$ with the same bin observation, corollary 2.5 gives, in the dimensionless scalar convention of this paper,

$$1 \leq \frac{\mathcal{Q}_{\text{bulk}}(M, \Delta)}{\chi \mathcal{Q}(M, \Delta)} \leq 1 + CM^{1/5} \log(1/M) \quad (159)$$

for small M , uniformly over the partitions under consideration. The assumptions are those of theorem 2.4, including fixed positive bulk transverse diffusion and nonzero mean speed. Mixing each selected field with the same vanishing uniform background preserves both its bin dependence and its exact budget. Thus every scalar equivalent and order above transfers to optimized flow-induced dispersion. In physical units the response factor $\ell_{\text{ref}}/k_{\text{ref}}$ in (7) must also be included. This is an optimal-value comparison, not a remainder estimate for every unmixed trial.

For one regular defect the uncertain root span is $w \simeq \Delta/\sqrt{a}$. Equation (142) places the local crossover at

$$\Delta \asymp m^{1/5} a^{3/10}. \quad (160)$$

The global mean is dominated by ordinary roots, whose curvatures give an integrable coefficient, so its threshold is $\Delta \asymp M^{1/5}$. For $N = 2^B$ equal bins, this means $B = (1/5) \log_2(1/M) + O(1)$ binary digits of offset resolution. In the sharp coarse regime, one further binary digit multiplies the leading response by $2^{-1/4}$. Refining a nested partition cannot worsen its optimum, since every coarser policy remains available; arbitrary nonnested partitions are not ordered by their widths alone.

These statements concern noiseless quantization followed by a static mobility choice. They do not optimize a measurement channel, assign a cost to information, or impose a fabrication length or mobility cap. The local value supplies an attained variational coefficient for the intermediate precision regime; the theorem does not require an explicit formula for its minimizing profile.

8 Numerical evidence and separate error checks

The analysis establishes continuum optimal values. The computations below serve two narrower purposes: they show how explicit designs behave at finite budgets and examine the local uncertainty-center problem numerically. They use deterministic offset quadrature, not sampling of tracer paths. No numerical value is assigned to $\mathcal{S}_q$, and no discretized calculation is used to prove the crossover theorem.

8.1 Circle responses and exact discrete budget accounting

By reflection symmetry we solve on $[0, \pi]$ with zero endpoint flux and double the integrated response. With n cells of width $h = \pi/n$, cell centers $s_i = (i+1/2)h$, and nonnegative face diffusivities $D_{i+1/2}$, the tridiagonal matrix is the finite-volume discretization of $-\partial_s(D\partial_s) + (c + \cos s)^2$. Its off-diagonal entry is $-D_{i+1/2}/h^2$; its diagonal is the sum of adjoining conductances plus the midpoint reaction. We solve against the vector of ones and approximate J_c by $2h \sum_i h_i$. There are no faces beyond the two endpoints. Every compared field, including the uniform field, is normalized to the same discrete budget

$$2h \sum_{i=1}^{n-1} D_i = M.$$

This normalization tends to the continuum integral constraint as the mesh is refined. It is necessary when sampling a singular or concentrated continuum shape; otherwise finite-grid comparisons would use slightly different resources.

The predetermined mean trial samples $|\sin s|^{-2/5}$. This integrable unrounded shape has the same leading mean coefficient as (78): it bounds the rounded profile from below by a multiplicative factor tending to one, so monotonicity and the sharp lower bound squeeze its leading cost to $K_1 M^{-1/4}$. For $q = 8/5$ and $q = 3$, we use the distance-based graded field (68) with (72), evaluated with its exact continuum normalization before imposing the discrete mass above. At criticality its leading coefficient is also K_{crit} : distance and $|\sin s|$ have ratio tending to one near either fold, their normalization integrals both equal $4 \log(1/R) + O(1)$, and the leading logarithmic contribution comes from those fold approaches. More explicitly, on the retained annuli $R \log(1/M) \leq r \leq r_*$, with fixed small $r_* > 0$, $(r + R)/(|\sin r| + R) = 1 + O(r_*^2)$. The part $r \geq r_*$ has critical moment $O(a_R^{-2/5})$, one logarithmic factor smaller than the leading contribution. The inner omitted annuli obey the same bounds as in the critical proof of theorem 4.1. First let $M \rightarrow 0$ and then $r_* \rightarrow 0$ to obtain the stated coefficient. The $q = 3$ field is an order-optimal trial; its scaled cost is not identified with $2^{(33-12)/7} \mathcal{S}_3$.

We also evaluate a simple exact-observation trial to illustrate finite-budget performance. For $0 \leq c < 1$, put $r = \arccos(-c)$, $a = 1 - c^2$, and $R = (40M/(3a))^{1/5}$. If $R < 0.3 \min(r, \pi - r)$, use the quadratic profile (112) centered at r on the half-wall. Otherwise use a constant patch adjacent to π of extent

$$E = \min\{\pi, \max(4M^{1/7}, 2(\pi - r))\}.$$

The full-wall field is its reflection; in the continuum the patch value is $M/(2E)$. For $c \geq 1$ use uniform mobility, and extend the policy to negative c by translation by π . The discrete normalization is applied to each realization. This is an evaluated trial, distinct from the policy used for the sharp asymptotic proof in section 6. Its finite-budget values do not certify the exact-observation optimum.

For these symmetric fields the ensemble moment is $(1/2) \int_0^2 J_c^q dc$. We subdivide this interval at 0, 0.5, 0.9, 1, 1.1, 1.5, 2 and at the valid points $1 \pm \lambda w$, with $\lambda = 0.1, 1, 10, 100$ and w drawn from the squared grading radius, $(M/(4\pi))^{1/3}$, and $M^{2/7}$. For the unrounded $q = 1$ trial the first of these subdivision widths is also set to $M^{2/7}$; it is a quadrature scale, not a profile cutoff. Duplicate and out-of-range points are removed. Each subinterval uses Gauss–Legendre quadrature. The main computations have $n = 32768$ and 48 nodes per subinterval. Both the full list of subdivisions and the unscaled ensemble moments are supplied with the data.

Figure 1 and table 2 show the finite-budget distinction. The predetermined trial is about 3.3% worse than uniform mobility at $M = 10^{-3}$, despite its smaller asymptotic coefficient: the limiting ratio is $K_1/[C_0^2(2\pi)^{1/4}/\sqrt{\pi}] = 0.9529814\dots$. At $M = 10^{-9}$ its numerical improvement is only about 2.1%. The observed trial yields a larger improvement, but its ratio to the predetermined trial is not

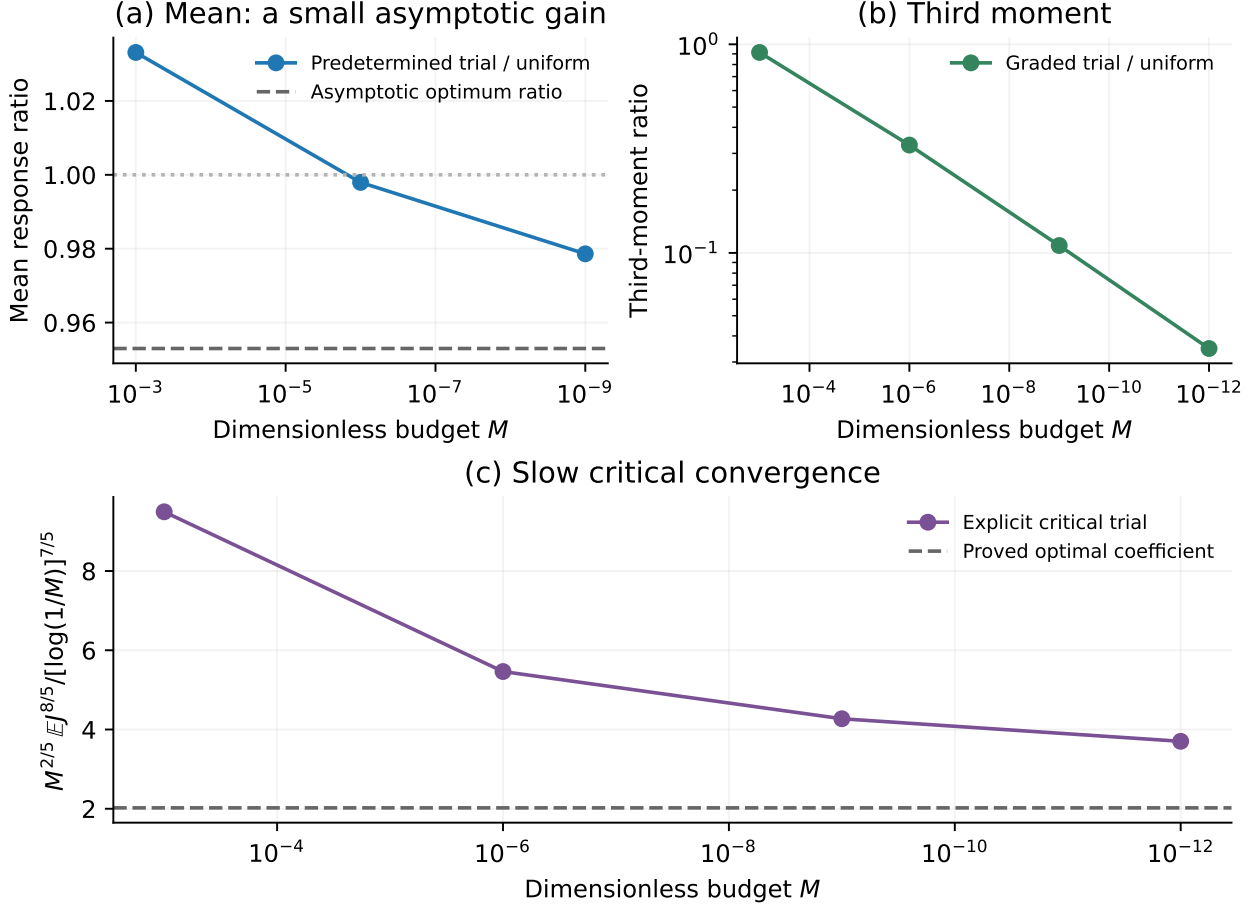


Figure 1: Finite-volume evaluation of explicit mobility trials, with the same discrete budget for each field. (a) A mean-optimal asymptotic shape can be worse than the uniform field at finite budget. (b) Grading has a larger effect on the third moment. (c) The critical scaled trial cost converges slowly toward the proved optimal coefficient. Curves are trial evaluations, not finite-budget optimal values; the dashed coefficients are continuum asymptotic results.

the ratio of their finite-budget optima. For the third moment the graded-to-uniform ratio decreases from about 0.914 at 10^{-3} to 0.0348 at 10^{-12} . At the critical order, the scaled trial moment decreases from 9.495 to 3.702 over that budget range, still well above $K_{\text{crit}} = 2.02233076 \dots$

We refine space and offset quadrature separately at the smallest budget for each of $q = 1, 8/5, 3$. Doubling the half-wall grid to 65536 cells at fixed 48-node quadrature changes the predetermined mean trial, critical moment trial, and third-moment trial by 0.0158%, 0.0084%, and 0.1184%, respectively, in absolute relative terms. Doubling the quadrature to 96 nodes at the original grid changes those three quantities by less than 10^{-8} relative. The discontinuous observed-trial branch selection gives a larger, but still small, quadrature change of 6.5×10^{-6} at $M = 10^{-9}$. A combined 65536-cell, 96-node calculation is also included. These checks assess stability of the reported trial values; they are not rigorous continuum error bounds or a test of the full bulk-surface model.

Table 2: Mean response ratios from the main circle computations. Both columns concern explicitly evaluated fields, with no numerical optimization over circle coefficients.

M	Predetermined trial / uniform	Observed trial / predetermined trial
10^{-3}	1.0331	0.8269
10^{-6}	0.9980	0.5942
10^{-9}	0.9786	0.4297

Table 3: Uncertain-center discrete values with 400 cells. Center quadrature has 96 nodes at $\eta = 2, 8, 192$ at $\eta = 32$, and one known center at zero. The lower value and gap use (162) for the same finite-dimensional objective.

η	Computed value	Discrete tangent lower value	Gap δ_h
0	6.222655	6.220971	0.0017
2	6.633100	6.630905	0.0022
8	8.082211	8.081774	0.00044
32	11.138582	11.138439	0.00014

8.2 The local uncertain-center problem

For (141), restrict mobility to $[-L, L]$ with $L = 6 + \eta/2$. Use n cells of width $h = 2L/n$ and optimize the nonnegative interior face masses $p_i = hD_i$, with $\sum_i p_i = 1$. The reaction at cell center x_j for a center z is $(x_j - z)^2$, and face conductance is p_i/h^3 . The interior problem has zero endpoint flux. Mobility is exactly zero outside the interval, so its exterior response is added analytically. Its averaged value is

$$\frac{2}{\eta} \log \frac{L + \eta/2}{L - \eta/2} \quad (\eta > 0), \quad \frac{2}{L} \quad (\eta = 0). \quad (161)$$

There is therefore no omitted reciprocal-potential tail; the remaining domain restriction is on where mobility can be placed. For $\eta > 0$ the center average uses Gauss–Legendre quadrature on $[-\eta/2, \eta/2]$.

We minimize this convex discrete objective $F_h(p)$ by sequential least squares programming, starting from uniform face masses and using the exact matrix derivative. If h_z denotes the solved source vector, that derivative is

$$g_i = \partial_{p_i} F_h = - \sum_z w_z \left(\frac{h_{z,i+1} - h_{z,i}}{h} \right)^2.$$

For any feasible returned vector p , convexity gives the global discrete bounds

$$F_h(p) - \delta_h \leq \inf_{r_i \geq 0, \sum r_i = 1} F_h(r) \leq F_h(p), \quad \delta_h = g \cdot p - \min_i g_i. \quad (162)$$

Negligible feasibility error is projected onto the simplex before evaluating these quantities. The formula is an analytical certificate for the discretized problem, evaluated in floating-point arithmetic. It neither bounds continuum discretization error nor provides interval-certified arithmetic.

Table 3 and figure 2 show the 400-cell calculations. At $\eta = 0$ the computed value 6.222655 lies slightly below the exact continuum value $C_{\text{pl}} = 6.222823736\dots$, illustrating why a discrete tangent

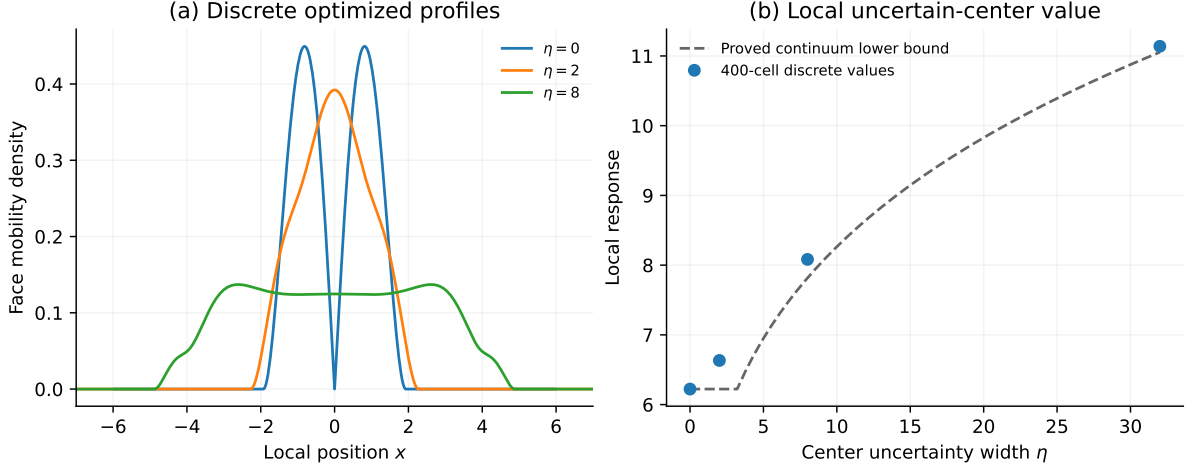


Figure 2: Finite-dimensional optimization of the local uncertain-center problem. (a) Optimized face profiles broaden as center uncertainty increases in the displayed examples. (b) Discrete computed values and the proved continuum bound $\max(C_{\text{pl}}, C_0\eta^{1/4})$. The discrete markers have discretization error and are not certified continuum upper bounds. Neither the figure nor the theorem asserts monotonicity of all minimizing profiles.

interval must not be read as a continuum interval. Increasing the grid from 200 to 400 cells changes the objective by about 0.0082%, 0.0136%, and 0.518% at $\eta = 0, 2, 32$, respectively. At $\eta = 0, 2$, these changes are smaller than the remaining discrete optimization uncertainty. They show stability of the returned objective values but do not separately resolve spatial discretization error. Re-evaluating every 400-cell field on 384 center nodes changes the objective by less than 10^{-12} relative. At $\eta = 32$, enlarging L from 22 to 33 while keeping $h = 0.11$ (400 to 600 cells) changes the computed value by 6.8×10^{-8} relative. This domain check concerns the computed value; its difference is smaller than the discrete optimization gaps and is not an independent rigorous tail bound.

A deliberately underresolved parameter calculation exposes a different error. At $\eta = 32$, optimizing 200 cells on only 24 center nodes gives 10.84053 with a small discrete tangent gap. Re-evaluating that same field on 384 nodes gives 12.09482, an increase of about 11.6%. A small optimization gap can therefore certify an overfitted quadrature problem. Spatial refinement, independent parameter quadrature, and optimization error must be checked separately. The continuum claims remain the exact variational characterizations and proved limits, not decimal values inferred from these finite grids.

8.3 Reproducibility

The accompanying `numerics/reproduce.py` regenerates the data, both figures, and the numerical table rows. The saved `data/numerical-evidence.json` includes every grid size, quadrature subdivision, feasibility check, optimizer status, tangent gap, and refinement used above. The script requires Python, NumPy, SciPy, and Matplotlib and imports the documented tridiagonal circle solver and graded trial construction from the repository's `scripts` directory. The folder README gives the exact commands and environment versions. Figures and tables are included in the manuscript source so that LaTeX compilation does not require rerunning the numerical optimization.

9 Discussion

The results distinguish three effects of a mobility budget. Uniform mobility regularizes each kinetic defect on a local diffusion length. Predetermined grading changes how the budget is shared among possible defect positions, delaying the disorder-moment transition from $4/3$ to $8/5$. Observation permits placement at the realized defects and changes the optimal mean exponent from $1/4$ to $1/5$. The finite-bin theorem connects these decisions quantitatively: the local uncertainty must be compared with the placement width, and the resulting conditional values must still be integrated through the fold configurations.

The strongest coefficients are not all elementary numbers. Above the predetermined threshold, $\mathcal{S}_q$ is an attained whole-line integrated fold-response minimum. At intermediate precision, $\mathcal{F}_{\text{ctr}}$ is an attained uncertain-center minimum and $\mathcal{H}$ is its explicit integrable average. The compactness and recovery arguments prove exact asymptotic values in both cases. They require neither an explicit minimizing density nor uniqueness. For the predetermined supercritical value, the sharp lower bound requires equal limiting allocations to the two rescaled folds and rules out wasted limiting mass; achieving only the optimal order does not impose this allocation.

These distinctions matter when using the results as design guidance. The mean improvement from predetermined grading is small in its leading coefficient, and a trial motivated by that limit can perform worse than uniform mobility at a finite budget. The critical logarithm also converges slowly. Numerical trial costs therefore cannot be substituted for the finite-budget optimum, and an asymptotic coefficient does not by itself prescribe a practical coating at a given physical budget. The numerical resolution studies in [section 8](#) make the separate discretization limitation explicit.

The transport comparison holds at fixed positive transverse bulk diffusivity, constant positive adsorption affinity, nonzero mean speed, and a connected one-dimensional wall. It concerns stationary long-time flow-induced dispersion. Molecular axial diffusion contributes separately as in [\(17\)](#); under the stated boundedness conditions it does not change the divergent leading optima. The whole-line source models are local energy-dual limits and do not represent particles in equilibrium on an infinite wall. Likewise, disorder moments describe a collection of fixed channels, not high displacement moments or anomalous time dependence within one channel.

The admissible design class idealizes control of tangential diffusivity independently of affinity, desorption, and geometry. The analysis does not establish that these controls can be varied independently in a particular surface chemistry. The theorems concern the specified integral-budget class; they do not establish optimal values under additional pointwise, background, or fabrication constraints. A vanishing bulk diffusivity or a kinetic family that nearly vanishes everywhere can invalidate the uniform physical comparison; the Gaussian amplitude-collapse example shows why local zero geometry alone is insufficient. The generic-fold theorem requires its stated transverse and sampling assumptions and does not cover higher degeneracies, simultaneous folds, or unknown fold locations. Finally, equal-bin observation has no measurement cost or stochastic noise model. These boundaries specify the transport and information problem solved by the present laws.

References

- [1] Joe Alexandersen and Ole Sigmund. Revisiting the optimal thickness profile of cooling fins: A one-dimensional analytical study using optimality conditions. In *2021 20th IEEE Intersociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems (ITherm)*, pages 24–30. IEEE, 2021. doi: 10.1109/ITherm51669.2021.9503196. Accepted manuscript available from the [University of Southern Denmark](#).

- [2] Arthur Alexandre, Thomas Guérin, and David S. Dean. Generalized Taylor dispersion for translationally invariant microfluidic systems. *Physics of Fluids*, 33:082004, 2021. doi: 10.1063/5.0057584. URL <https://arxiv.org/abs/2105.06212>.
- [3] Amal Alphonse, Petar Kunštek, and Marko Vrdoljak. Optimal design with uncertainties: a risk-averse approach, 2026. URL <https://arxiv.org/abs/2602.19869>.
- [4] Jean-Marc Azaïs and Mario Wschebor. *Level Sets and Extrema of Random Processes and Fields*. Wiley, Hoboken, NJ, 2009. doi: 10.1002/9780470434642. URL <https://www.math.univ-toulouse.fr/~azais/styles/other/student/level.pdf>. Author-hosted book draft available at the linked URL.
- [5] M. V. Berry, J. P. Keating, and H. Schomerus. Universal twinkling exponents for spectral fluctuations associated with mixed chaology. *Proceedings of the Royal Society of London A*, 456:1659–1668, 2000. doi: 10.1098/rspa.2000.0580. URL <https://michaelberryphysics.wordpress.com/wp-content/uploads/2013/07/berry320.pdf>.
- [6] Guy Bouchitté and Giuseppe Buttazzo. Characterization of optimal shapes and masses through Monge–Kantorovich equation. *Journal of the European Mathematical Society*, 3(2):139–168, 2001. doi: 10.1007/s100970000027. URL <https://ems.press/journals/jems/articles/123>.
- [7] Giuseppe Buttazzo and Faustino Maestre. Optimal shape for elliptic problems with random perturbations. *Discrete and Continuous Dynamical Systems*, 31(4):1115–1128, 2011. doi: 10.3934/dcds.2011.31.1115. URL <https://arxiv.org/abs/1002.2770>.
- [8] Giuseppe Buttazzo, Edouard Oudet, and Bozhidar Velichkov. A free boundary problem arising in PDE optimization. *Calculus of Variations and Partial Differential Equations*, 54(4):3829–3856, 2015. doi: 10.1007/s00526-015-0923-1. URL <https://arxiv.org/abs/1506.00141>. The cited Proposition 4.1 is in the openly accessible preprint.
- [9] Sergio Conti, Harald Held, Martin Pach, Martin Rumpf, and Rüdiger Schultz. Shape optimization under uncertainty—a stochastic programming perspective. *SIAM Journal on Optimization*, 19(4):1610–1632, 2009. doi: 10.1137/070702059.
- [10] Loren H. Dill and Howard Brenner. A general theory of Taylor dispersion phenomena: III. surface transport. *Journal of Colloid and Interface Science*, 85(1):101–117, 1982. doi: 10.1016/0021-9797(82)90239-9.
- [11] Masatoshi Fukushima, Yoichi Oshima, and Masayoshi Takeda. *Dirichlet Forms and Symmetric Markov Processes*, volume 19 of *De Gruyter Studies in Mathematics*. De Gruyter, Berlin, second revised and extended edition, 2011. doi: 10.1515/9783110218091.
- [12] Pierre Grisvard. *Elliptic Problems in Nonsmooth Domains*, volume 69 of *Classics in Applied Mathematics*. Society for Industrial and Applied Mathematics, Philadelphia, 2011. doi: 10.1137/1.9781611972030. Reprint of the 1985 edition.
- [13] Maximilien Levesque, Olivier Bénichou, Raphaël Voituriez, and Benjamin Rotenberg. Taylor dispersion with adsorption and desorption. *Physical Review E*, 86:036316, 2012. doi: 10.1103/PhysRevE.86.036316. URL <https://arxiv.org/abs/1211.5224>.
- [14] M. E. J. Newman, Michelle Girvan, and J. Dooyne Farmer. Optimal design, robustness, and risk aversion. *Physical Review Letters*, 89:028301, 2002. doi: 10.1103/PhysRevLett.89.028301. URL <https://arxiv.org/abs/cond-mat/0202330>.

- [15] Naci Saldi, Serdar Yüksel, and Tamás Linder. Finite model approximations and asymptotic optimality of quantized policies in decentralized stochastic control. *IEEE Transactions on Automatic Control*, 62(5):2360–2373, 2017. doi: 10.1109/TAC.2016.2613902. URL <https://mast.queensu.ca/~linder/pdf/SaYuLi17.pdf>.
- [16] Serdar Yüksel and Tamás Linder. Optimization and convergence of observation channels in stochastic control. *SIAM Journal on Control and Optimization*, 50(2):864–887, 2012. doi: 10.1137/100808976. URL <https://arxiv.org/abs/1009.3824>.